

**Beautiful Dua
from Quran**

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

Fall

(2024)



Q. What is the type of hybridization in Nickel in $\text{Ni}(\text{CO})_4$ and $[\text{Ni}(\text{CN})_4]^{2-}$

[Haq Nawaz Bhatti Page No. 157 onward](#)

[Sanaullah Page No. 206 and 208](#)

Q. Metals do not form ionic or covalent bonds?

Metals, in their pure or elemental form, do not form ionic or covalent bonds. Instead, they form **metallic bonds**, which are a special type of chemical bond. In metallic bonding, each metal atom

releases its outer (valence) electrons, and these electrons move freely among all the atoms. This creates a "sea of electrons" that surrounds positively charged metal ions (atoms that have lost electrons).

This unique bonding structure is responsible for the characteristic properties of metals. The freely moving electrons allow metals to conduct electricity and heat. Also, this bonding lets metal layers slide over each other without breaking, which makes metals **malleable** (easily shaped or bent) and **ductile** (can be drawn into wires).

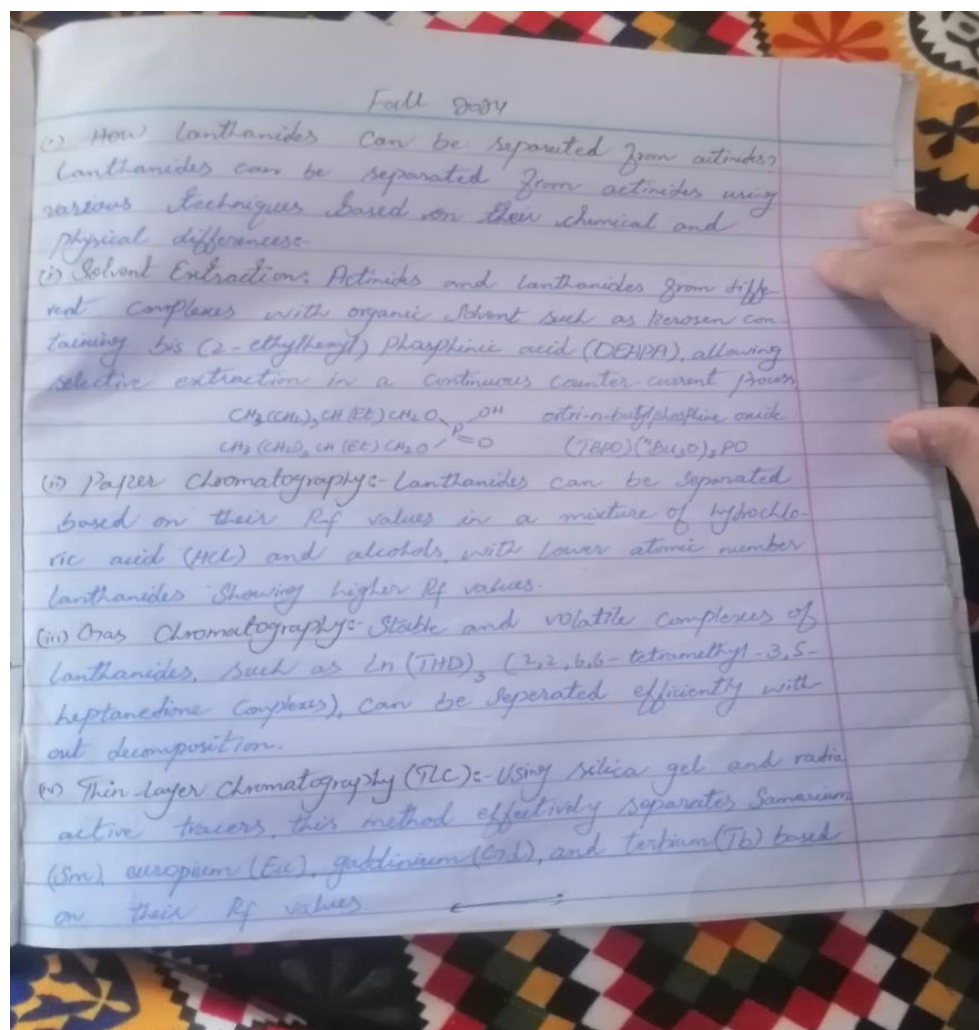
Example of Metallic Bonding (in Pure Sodium – Na):

In pure sodium metal, each sodium (Na) atom **loses one valence electron**, which becomes part of the "sea of electrons." The remaining atom becomes a **positive ion (Na⁺)**.



These Na⁺ ions are arranged in a regular pattern (lattice), and the **free-moving electrons** hold the entire structure together. This is why sodium metal can **conduct electricity** and is **malleable** and **ductile**.

Q. How lanthanides can be separated from actinides?



Q. Neither La^{3+} nor Lu^{3+} ion show any colour?why

The La^{3+} and Lu^{3+} ions are colorless because they have no electrons in their 4f orbitals, resulting in no f-f electron transitions that could absorb visible light and produce color.

- ❖ **La^3 (Lanthanum):** Lanthanum's electron configuration is $[\text{Xe}] 5d^1 6s^2$. When it forms La^3 , it loses the $5d^1$ and $6s^2$ electrons, leaving an electron configuration of $[\text{Xe}]$, which has a completely empty 4f orbital ($4f^0$). With no electrons in the 4f orbitals, there are no f-f transitions possible, and thus, La^{3+} does not absorb visible light, appearing colorless.
- ❖ **Lu^3 (Lutetium):** Lutetium's electron configuration is $[\text{Xe}] 4f^{14} 5d^1 6s^2$. Upon forming Lu^3 , it loses the $5d^1$ and $6s^2$ electrons, resulting in $[\text{Xe}] 4f^{14}$. Here, the 4f orbitals are completely filled. Since f-f transitions require partially filled orbitals to allow electron movement between energy levels, a fully occupied $4f^{14}$ configuration cannot undergo such transitions, making Lu^3 colorless as well.

In lanthanide ions, color typically arises from f-f electronic transitions, which occur when 4f orbitals are partially filled (e.g., in ions like Ce^3 or Nd^3). Since La^3 has an empty 4f orbital and Lu^3 has a fully filled 4f orbital, neither can produce these transitions, resulting in no absorption in the visible spectrum and, consequently, no color.

Q.What are the toxic effects of actinides?

Actinides are a group of 15 radioactive metallic elements from Actinium (Ac) to Lawrencium (Lr) with atomic numbers 89 to 103. These elements are both radioactive and chemically toxic. When they enter the human body through inhalation, ingestion, or wounds, they can cause severe health issues due to radiation and heavy metal toxicity.

1. Radiological Toxicity:

Actinides emit alpha, beta, and gamma radiation. Alpha particles are highly damaging when emitted inside the body and can cause cell death and DNA mutations.

Example: Inhalation of Plutonium-239 (Pu-239) particles leads to lung cancer because alpha particles damage lung tissues.

Example: Americium-241 (Am-241) emits both alpha and gamma rays and accumulates in the bone marrow, increasing the risk of leukemia.

2. Chemical Toxicity:

As heavy metals, actinides disrupt biological functions by mimicking essential metals like calcium and iron, leading to organ damage.

Example: Uranium (U-238) gets deposited in bones and kidneys, causing nephrotoxicity (kidney damage) and bone marrow suppression.

Example: Neptunium-237 (Np-237) affects liver and immune function, especially in soluble chemical forms.

3. Environmental and Exposure Pathways:

Actinides enter the body through exposure to nuclear materials, waste, or weapons. They persist in the environment and can contaminate air, water, and soil.

Example: Depleted Uranium (DU) used in ammunition produces fine dust particles that cause genetic mutations and lung problems when inhaled.

Example: Curium (Cm), Californium (Cf), and Berkelium (Bk) are synthetic actinides that can spread through groundwater and pose long-term radiological risks.

Q.What is used as coolant in nuclear reactors?

[Haq Nawaz Bhatti Page No. 113](#)

Q.Which element is used as an energy source in heart pacemakers?

A **heart pacemaker** is a small electronic medical device that is **implanted inside the chest** to help control **abnormal heart rhythms** (arrhythmias). It sends **electrical impulses** to the heart muscles to ensure that the heart beats at a **normal and steady rate**. Pacemakers are often used in patients who suffer from **bradycardia** (slow heart rate) or **heart block**, where the natural electrical system of the heart is damaged or weakened. The device helps improve **blood circulation**, prevent **fatigue**, and reduce the risk of **heart failure or cardiac arrest**.

A special radioactive element called **Plutonium-238 (Pu-238)** was used as the **energy source** in some heart pacemakers. Plutonium-238 is a **radioactive isotope** that emits **alpha radiation**, which is not highly penetrating but releases a steady amount of **heat energy**. This heat was converted into **electricity** using a **thermoelectric generator** inside the pacemaker, ensuring a **long-lasting power supply**. Due to its **long half-life of 87.7 years**, Pu-238-powered pacemakers could operate for several decades without needing a battery replacement — making them ideal for patients who were not good candidates for repeat surgeries.

However, due to concerns about **radiation exposure**, **cost**, and the availability of safer alternatives, the use of Pu-238 has been largely discontinued in modern medical practice. Today, most pacemakers use **lithium-based batteries**, which are **compact, safe, and last between 8 to 15 years**. Although Pu-238-powered pacemakers are now rare, they were a remarkable use of **nuclear technology in life-saving medical applications**.

Q.What is misch metal? What are its uses?

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Mischmetal is an alloy made from a mixture of rare-earth elements. It is obtained from minerals like monazite and bastnasite and is silvery and brittle in appearance. It is non-radioactive and used in various industrial applications.

Typical composition includes:

- ❖ 50–55% Cerium (Ce)
- ❖ 20–25% Lanthanum (La)

- ❖ 15–18% Neodymium (Nd)
- ❖ 4–5% Praseodymium (Pr)
- ❖ ~5% Other lanthanides + small amounts of Iron (Fe)

Uses of Mischmetal:

1. Flint for Lighters:

Mischmetal is used in making ferrocerium for lighter flints. Cerium in the alloy produces sparks when scraped. It is used in cigarette lighters, welding torches, and fire starters.

2. Metallurgical Deoxidizer:

In steel and iron industries, mischmetal is used to remove oxygen and sulfur. This improves metal quality and strength. It is especially useful in casting and refining processes.

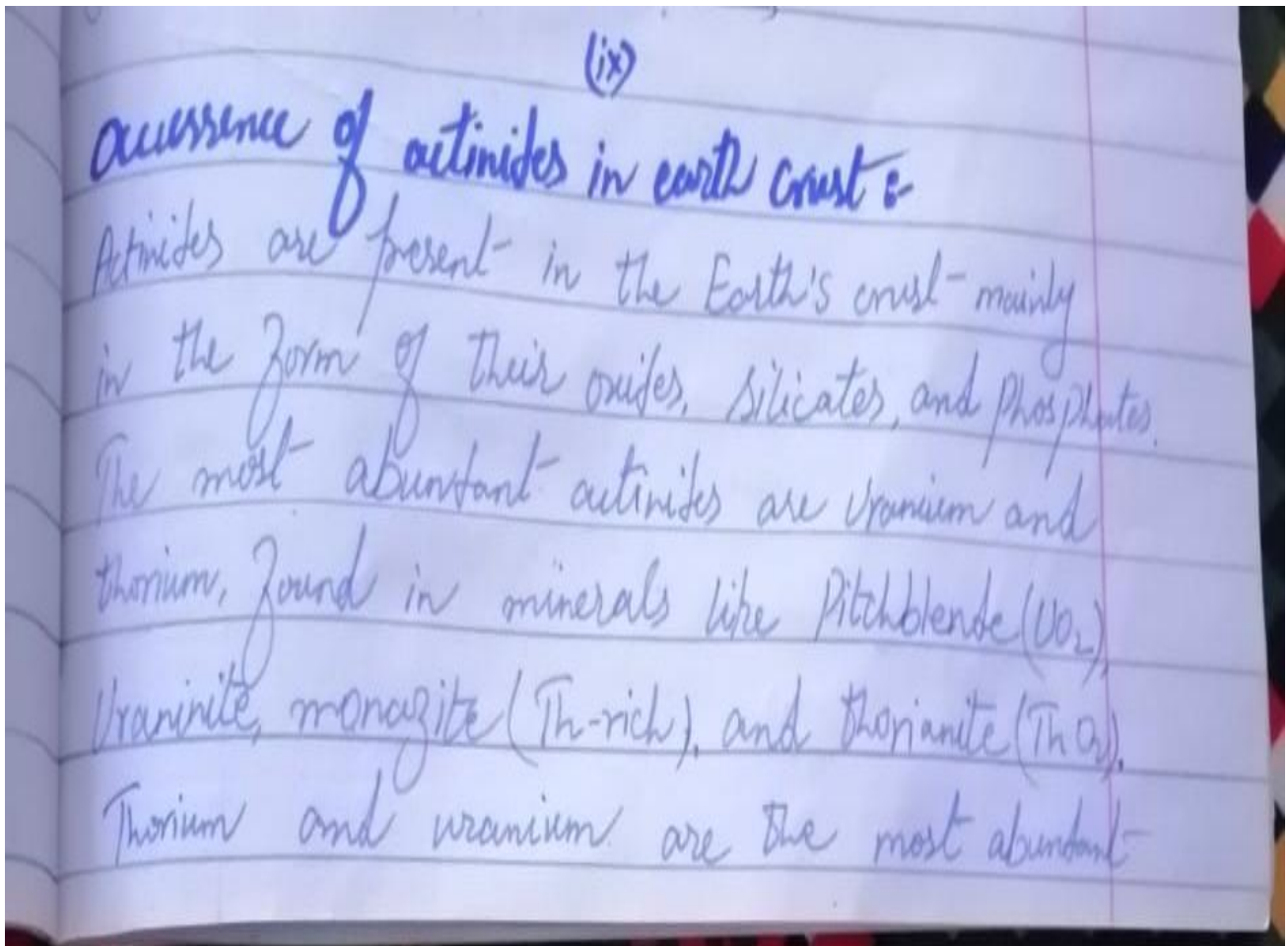
3. Nickel-Metal Hydride (Ni-MH) Batteries:

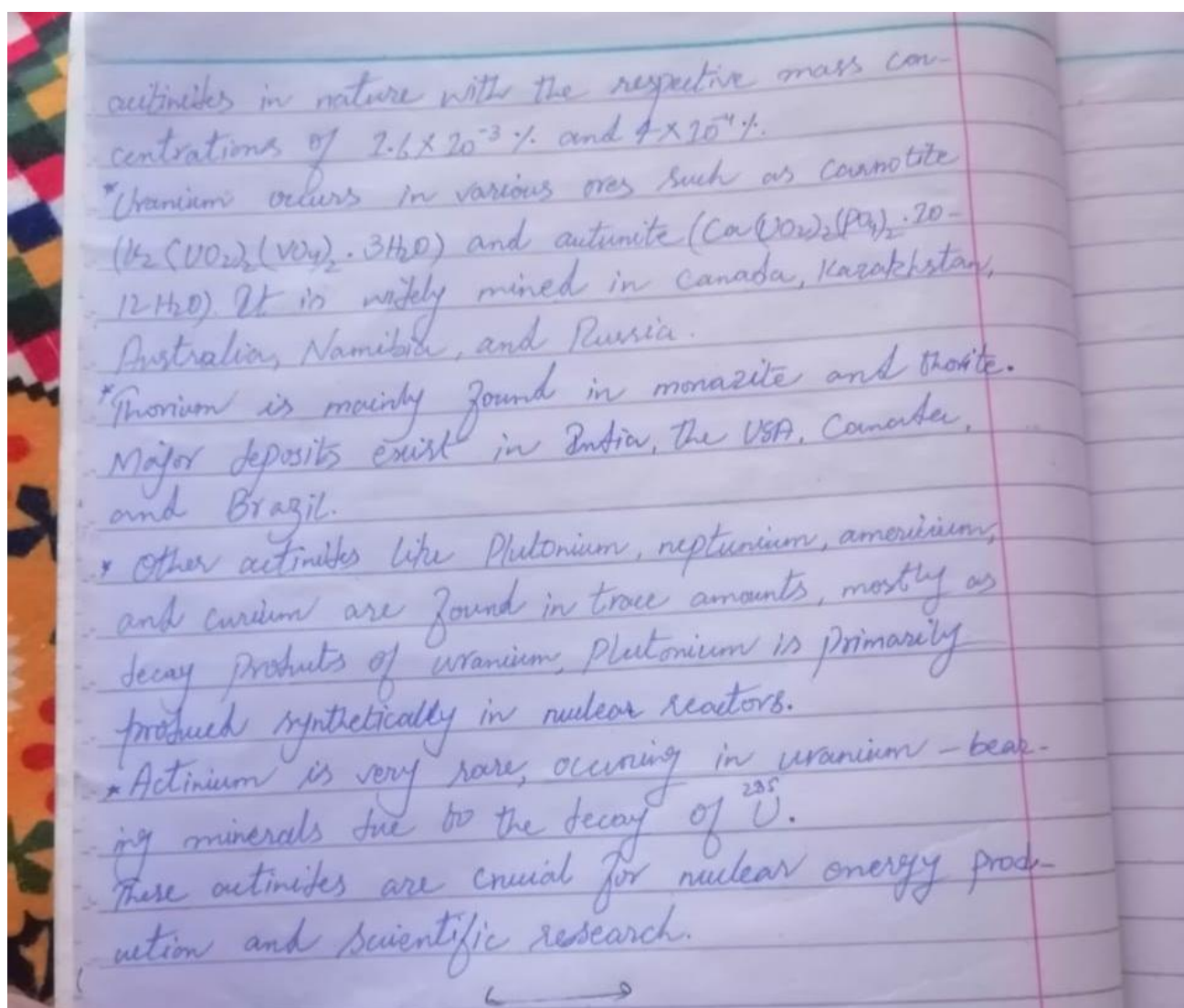
Mischmetal is a component in hydrogen-storing alloys used in batteries. These batteries are found in hybrid cars and electronics. It provides better energy storage and recharge cycles.

What is hybridization in AB₉ molecule?

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Q. Briefly write the occurrence of actinides in earth crust?





Q.What is the effective nuclear charge experience by a valence p- electron in boron?

Effective Nuclear Charge (Z_{eff}) is the net positive charge experienced by an electron in a multi-electron atom. Due to the shielding effect of inner electrons, the outer (valence) electrons do not feel the full attractive force of the nucleus.

Formula:

$$Z_{\text{eff}} = Z - S$$

Where:

- ❖ Z = Atomic number of the element
- ❖ S = Shielding or screening constant (estimated using Slater's Rules)

For Boron (B):

- ❖ Atomic number, $Z=5$
- ❖ Electron configuration = $1s^2 2s^2 2p^1$
- ❖ We are calculating Z_{eff} for a 2p electron (valence p-electron).

Now apply Slater's Rules to find S:

1. Group electrons:

$(1s^2)$
 $(2s^2, 2p^1) \leftarrow$ group of interest

2. Apply Slater's Rules for 2p electron:

- ❖ Electrons in the same group (2s and 2p): each contributes 0.35
 $\rightarrow 2 \text{ electrons} \times 0.35 = 0.70$
- ❖ Electrons in the $n=1$ shell ($1s^2$): each contributes 0.85
 $\rightarrow 2 \text{ electrons} \times 0.85 = 1.70$
- ❖ Total $S = 0.70 + 1.70 = 2.40$

$$Z_{\text{eff}} = Z - S = 5 - 2.40 = 2.60$$

Read page number 8 Haq nawaz bhatti

Q.Explain the magnetic behavior of O_2 molecule by comparing its VBT and MOT pictures?

The magnetic behavior of the oxygen molecule (O_2) can be explained by comparing the Valence Bond Theory (VBT) and Molecular Orbital Theory (MOT), which differ notably in their treatment of electron pairing and magnetism.

✔ Valence Bond Theory (VBT) Picture of O_2

Each oxygen atom has the electronic configuration $1s^2 2s^2 2p^4$.

VBT says two oxygen atoms form a double bond:

- ❖ One sigma (σ) bond (head-on overlap of orbitals)
- ❖ One pi (π) bond (side-by-side overlap of p orbitals)

Electron Pairing:

VBT assumes all electrons involved in bonding are paired.

Bond Order:

Bond order = 2 (one σ + one π bond)

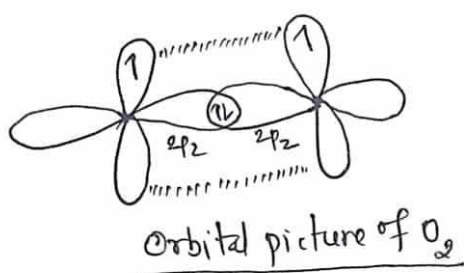
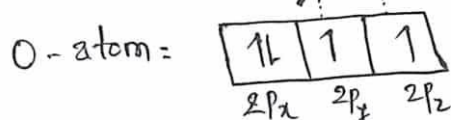
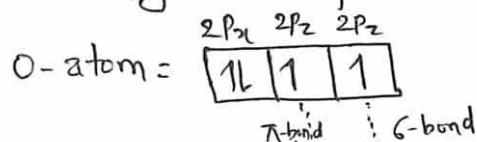
Magnetic Behavior:

Since all electrons are paired, VBT predicts O_2 to be diamagnetic (not attracted to magnetic fields).

Limitation:

This contradicts experimental observations. Oxygen is actually paramagnetic, which VBT fails to explain.

Electronic configuration of O-atom = $1s^2 2s^2 2p_x^2 2p_y^1 2p_z^1$

**Molecular Orbital Theory (MOT) Picture of O_2**

MOT treats electrons as occupying molecular orbitals that are spread over the entire molecule, not localized between atoms.

Total Electrons in O_2 :

8 electrons from each oxygen = 16 electrons

- Molecular Orbital Filling Order:

$$\sigma(1s)^2 \sigma^*(1s)^2 \sigma(2s)^2 \sigma^*(2s)^2 \sigma(2p_z)^2 \pi(2p_x)^2 = \pi(2p_y)^2 \pi^*(2p_x)^1 \pi^*(2p_y)^1$$

- Bond Order Calculation:

$$\text{Bond order} = \frac{N_b - N_a}{2} = \frac{10 - 6}{2} = 2$$

(Where N_b = bonding electrons, N_a = antibonding electrons)

Electron Pairing:

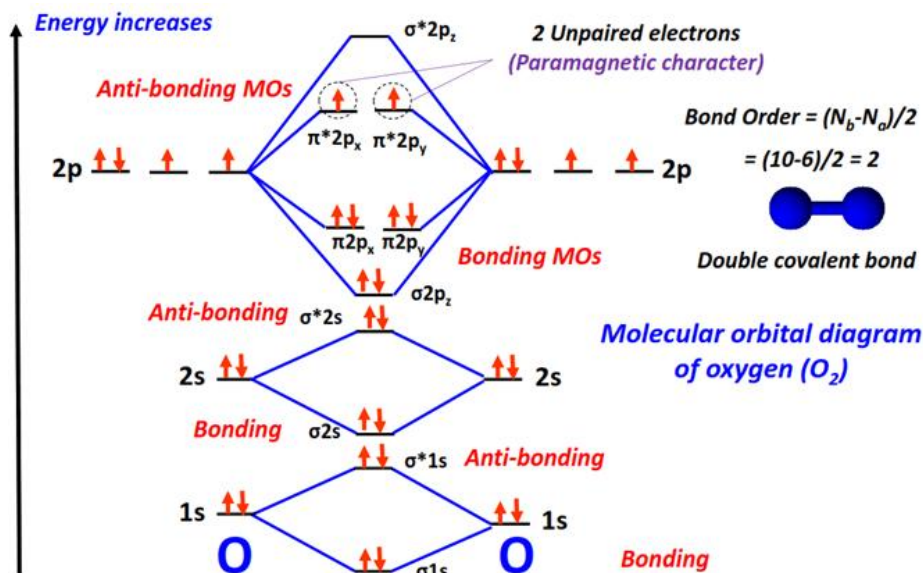
Two electrons are unpaired in the antibonding $\pi^* 2p$ orbitals.

Magnetic Behavior:

Because of these two unpaired electrons, MOT correctly predicts that O₂ is paramagnetic (attracted to magnetic fields).

Experimental Evidence:

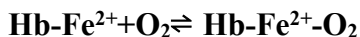
Liquid oxygen is suspended between the poles of a magnet—clear proof of paramagnetism, explained only by MOT.



Q. Give two applications of metal complexes in biological systems?

1. Oxygen Transport – Hemoglobin (Iron Complex)

One important application of metal complexes in biological systems is **oxygen transport** in the human body. The protein **hemoglobin**, found in red blood cells, contains an **iron(II) (Fe²⁺) complex** at the center of a **heme group** (a porphyrin ligand). This metal complex binds oxygen molecules reversibly. The reaction involved can be represented as:

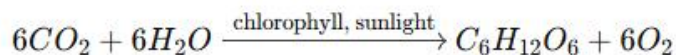


In this reaction, the **Fe²⁺** ion binds with O₂ to form **oxyhemoglobin**. This allows oxygen to be transported from the lungs to body tissues, where oxygen is released for cellular use.

2. Photosynthesis – Chlorophyll (Magnesium Complex)

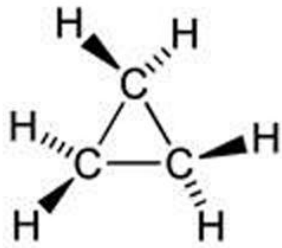
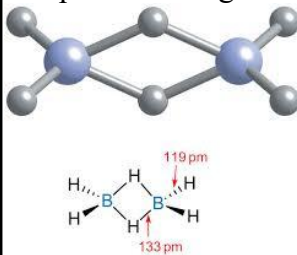
Another important application is in photosynthesis, where the metal complex chlorophyll plays a central role. Chlorophyll is a green pigment in plants that contains magnesium (Mg²⁺) at the center of a porphyrin-like ring. It captures solar energy to convert carbon dioxide and water into glucose and oxygen.

The overall reaction is:



Here, the Mg^{2+} ion helps in the absorption of light energy, making photosynthesis possible. This process is vital for producing food and releasing oxygen into the atmosphere.

Q.What is the difference between bent bond and bridge bond?

Feature	Bent Bond	Bridge Bond
Definition	A bent bond is a type of covalent bond formed when two atomic orbitals overlap at an angle rather than in a straight line, resulting in a curved or off-axis bond.	A bridge bond is a type of covalent bond where one atom (usually hydrogen) forms a bond with two other atoms simultaneously, creating a three-center bond.
Shape	The shape of a bent bond is curved or banana-like due to the angular overlap of orbitals, which deviates from the usual straight bond geometry.	In a bridge bond, the bridging atom is located between two atoms, giving the appearance of a bridge connecting them.
Example	Bent bonds are found in molecules like cyclopropane , where the carbon-carbon bonds are bent due to ring strain and reduced bond angles. 	Bridge bonds are observed in molecules like diborane (B₂H₆) , where hydrogen atoms bridge two boron atoms to complete bonding. 
Bond Type	A bent bond is a 2-center 2-electron (2c–2e) bond, meaning two atoms share two electrons in a distorted bonding arrangement.	A bridge bond is a 3-center 2-electron (3c–2e) bond, in which three atoms share two electrons, common in electron-deficient compounds.
Electron Sharing	In bent bonds, electrons are shared between two atoms only, but the overlap is not linear due to molecular geometry or strain.	In bridge bonds, the electrons are shared among three atoms, allowing bonding in molecules that do not have enough electrons for standard bonding.
Cause	Bent bonds are typically caused by orbital strain or forced small bond angles that prevent the orbitals from aligning in a straight line.	Bridge bonds arise mainly due to electron deficiency, especially in elements like boron and aluminum, where normal 2c–2e bonding is insufficient.

Q. Give any method of preparation of metal complexes?

A number of preparative techniques have been used to prepare metal carbonyls, and four general ones will be discussed here.

1. Direct Synthesis

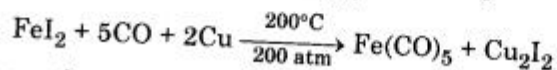
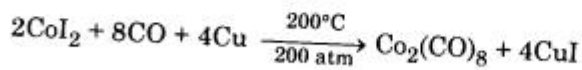
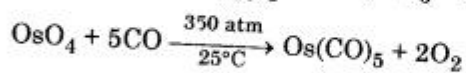
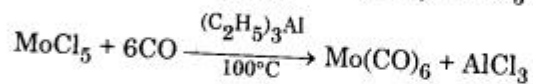
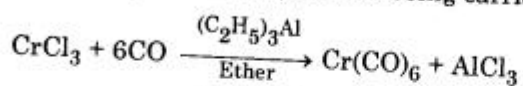
Carbon monoxide reacts with several finally divided metals to give metal carbonyls. Nickel, iron and cobalt carbonyls can be prepared in this way.



For most other metals, high temperatures and pressures are required to produce the metal carbonyls. By this direct reaction, $\text{Co}_2(\text{CO})_8$, $\text{Mo}(\text{CO})_6$, $\text{Ru}_4(\text{CO})_{12}$ and $\text{W}(\text{CO})_6$ have been prepared when suitable conditions are used.

2. Reductive Carbonylation

This type of reaction involves reducing a metal compound in the presence of CO. The reducing agents may include a variety of materials like Mg, Ag, Cu, Na, H_2 , LiAlH_4 depending on the particular synthesis being carried out.



Yields are often very poor in this process.

Long Question

Q.What is crystal field splitting? Give an account of the important factors which influence the magnitude of crystal field splitting.

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Q.Write optical Isomerism of octahedral complexes?

Page number 323 onward Haq Nawaz Bhatti

Q.How are lanthanides separated from different ores?

Page number 452 onward Haq Nawaz Bhatti

Q.How transuranic actinides were synthesized by artificial transmutation reactions?

Page number 480 onward Haq Nawaz Bhatti

Q.Discuss the John-Teller distortion theorem with examples?

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Q.Write a note on acid opening method of monazite ore for Ln-extraction?

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(2024)



Q.What is the role of stability in coordination complexes?

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OR

<https://www.slideshare.net/slideshow/stability-of-co-ordination-complexes-236855156/236855156>

Stability of Co-ordination Complexes



The stability of Co-ordination complexes is determined to be its existence in aqueous solution w.r.t its bond dissociation energy, Gibbs free energy (ΔG^0), standard electrode potential (E^0) or pH of the solution. Stability of complexes is also depend on the activation Energy (E_a) or speed of the substitution reaction.



(i) Thermodynamic Stability

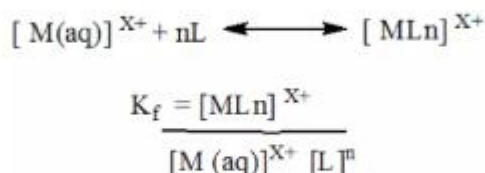
(ii) Kinetic Stability

Thermodynamic stability measure in term of complex to exist in equilibrium condition or its measure a tendency of a metal ion to form a particular complex ion.

- M-L bond energy
- Stability constant or formation constant (K_f)
- is directly related to ΔG^0

$$\begin{aligned}\Delta G^0_{\text{reaction}} &= \Delta G^0_{\text{product}} - \Delta G^0_{\text{reactant}} \\ \Delta G^0 &= -RT \ln K_f \\ \Delta G^0 &= -nFE^0 \\ RT \ln K_f &= nFE^0 \\ K_f &\propto E^0\end{aligned}$$

ΔG^0 is thermodynamic Property Hence formation Constant is measure as thermodynamic stability. Higher -Ve value of ΔG higher indicate that the complex formation is more stable and spontaneous in nature.



Thermodynamic stability is typically expressed in term of an overall formation or stability constant i.e, K_f

➤ The magnitude of formation constant is measure the stability of a complex.

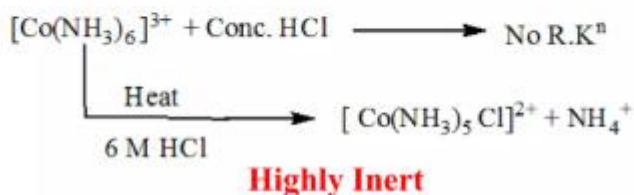
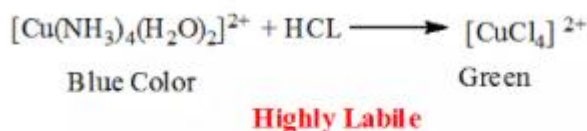
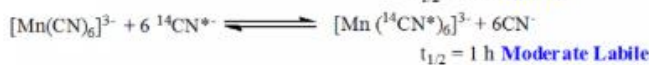
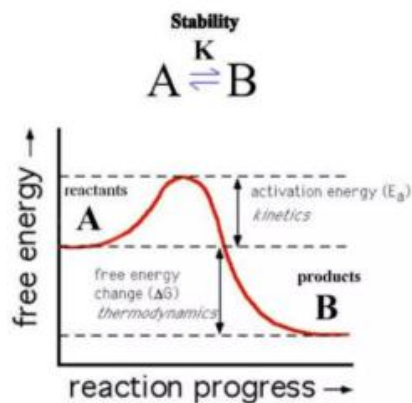
Kinetic Stability

The kinetic stability depends on the speed of the reaction, which depend on activation energy (E_a) and rate of the reaction.

1. Kinetically Inert
2. Kinetically Labile

Inert complex gives substitution reaction with difficulties $t_{1/2} > 1 \text{ min}$

Labile complexes gives substitution reaction rapidly $t_{1/2} < 1 \text{ min}$



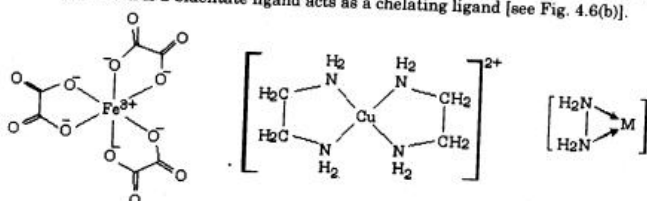
Q.What are metal chelates?

4.7 CHELATING LIGANDS AND CHELATES

A chelating ligand is a bidentate or polydentate ligand which is attached to the same central metal atom by two or more of its donor atoms resulting in the formation of a complex having a strain-free ring structure. The complex having the ring structure is called chelate or chelated complex. The chelate is also called by various other names like *cyclic complex*, *ring-type complex* etc. The formation of a chelate is called *chelation* or *cyclisation*.

Examples of the formation of chelates

- (i) When two molecules of ethylenediamine, $\text{NH}_2 - \text{CH}_2 - \text{CH}_2 - \text{NH}_2$ (en) which is a bidentate ligand, get attached with one Cu^{2+} ion through its two N-donor atoms of each molecule, complex ion viz. $[\text{Cu}(\text{en})_2]^{2+}$ or $[\text{Cu}(\text{NH}_2 - \text{CH}_2 - \text{CH}_2 - \text{NH}_2)_2]^{2+}$ which contains two 5-membered rings is obtained. In the formation of this complex ion, ethylenediamine molecule (bidentate ligand) acts as a chelating ligand and the complex ion thus formed is called a chelate [see Fig. 4.6 (a)].
- (ii) When three oxalate ions, $\text{C}_2\text{O}_4^{2-}$ or ox^{2-} get attached with one Fe^{3+} ion, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ or $[\text{Fe}(\text{ox})_3]^{3-}$ ion is obtained. This complex ion is a chelate. This chelate has three 5-membered rings. In the formation of this chelate $\text{C}_2\text{O}_4^{2-}$ ion which is a bidentate ligand acts as a chelating ligand [see Fig. 4.6(b)].



Q.What is the position of f-block elements in Periodic table?

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"f-block elements are those elements in which the differentiating electron enters the (n-2)f orbital. These elements are also called inner transition elements and are placed separately at the bottom of the periodic table in two horizontal rows to maintain its structure."

They are part of Group 3 and are divided into two series:

1. Lanthanide Series:

The lanthanide series includes elements with atomic numbers from 58 (Cerium) to 71 (Lutetium). In these elements, the differentiating electron enters the 4f-orbital, and their general electronic configuration is $Xe\ 4f^{1-14}\ 5d^{0-1}\ 6s^2$. These elements are commonly known as rare earth elements. Lanthanides mostly exhibit the +3 oxidation state, although some may show +2 or +4 as well. One of the key features of this series is the lanthanide contraction, which refers to the gradual decrease in atomic and ionic sizes across the series due to poor shielding by 4f electrons. Lanthanides also form coloured ions and are usually paramagnetic, a result of the presence of unpaired electrons in their f-orbitals.

2. Actinide Series:

The actinide series consists of elements with atomic numbers from 90 (Thorium) to 103 (Lawrencium). In these elements, the differentiating electron enters the 5f-orbital, and their general configuration is $Rn\ 5f^{1-14}\ 6d^{0-1}\ 7s^2$. Most actinides are radioactive, and several of them are transuranic elements, meaning they come after uranium in the periodic table. Actinides typically exhibit the +3 oxidation state, but they can also show oxidation states from +4 to +7, depending on the element and compound. Similar to the lanthanides, actinides experience actinide contraction, caused by poor shielding of 5f electrons. Because of their radioactivity and ability to release large amounts of energy, actinides are widely used in nuclear reactors and the production of nuclear weapons.

Q.Give names of some Lanthanide ores, from which it can be extracted?

Repeat in Long Question, Fall 2024

Q.What are the uses of Uranium other than nuclear fuel?

Although uranium is best known for its use as a nuclear fuel, it also has several other important uses, especially in industry, science, and military. These include:

1. Manufacturing of Armor and Ammunition:

- ❖ **Depleted uranium (DU)**, which is less radioactive than natural uranium, is used in the production of armor-piercing bullets and shells.
- ❖ It is also used to strengthen military vehicle armor due to its high density and hardness.

2. Use in Aviation and Ship Ballast:

- ❖ Because of its very high density, uranium is used as ballast in airplanes, missiles, and ships to help maintain balance and stability.

3. Production of Colored Glass and Ceramics:

- ❖ Uranium compounds (like uranium oxide) were historically used to make uranium glass (which has a greenish-yellow color and glows under UV light).
- ❖ It is also used as a coloring agent in ceramics and glazes.

4. Scientific Research and Radiography:

- ❖ Uranium is used in radiographic testing to check for cracks and flaws in metal parts (non-destructive testing).
- ❖ In science, it is used as a target material in particle accelerators and source material for the production of other radioactive isotopes.

5. Dating of Ancient Rocks (Uranium-Lead Dating):

Uranium isotopes are used in radiometric dating techniques to estimate the age of rocks, fossils, and **Earth's crust** (Uranium-238 decays to lead-206).

Q.Explain the structure of Hexaammine Iron (II) chloride and Hexazamine copper (II) by VBT?

[Answer Available below , Past Paper 2022](#)

[This answer according Warner Theory Haq Nawaz Bhatti page number 233 onward](#)

Q.What is the commonly found actinides in earth crust?

[Haq Nawaz Bhatti page number 480 onward](#)

Commonly Found Actinides in Earth's Crust

The actinide series includes elements from atomic numbers 89 to 103, but only four actinides occur naturally in the Earth's crust: Thorium (Th), Uranium (U), Protactinium (Pa), and Actinium (Ac). The rest are artificially produced in nuclear reactors. These naturally occurring actinides are important due to their availability and applications in energy, defense, and research.

1. Thorium (Th)

Thorium (atomic number 90) is the most abundant natural actinide. It is commonly found in monazite sands, especially in regions like India and Brazil. It exists mainly as thorium oxide (ThO₂). Thorium is a fertile nuclear material, meaning it can be converted into fissile uranium-233 inside reactors, making it valuable for nuclear power in the future.

Applications of Thorium:

- ❖ Thorium is used in thorium-based nuclear reactors as an alternative to uranium fuel.
- ❖ It is used in gas mantles for portable lighting because thorium oxide glows brightly when heated.
- ❖ It is also used in high-temperature alloys for aerospace and industrial applications due to its strength and heat resistance.

2. Uranium (U)

Uranium (atomic number 92) is the best-known actinide and occurs in minerals like pitchblende (UO_2) and carnotite. It naturally exists as two main isotopes: U-238 (99.3%) and U-235 (0.7%). U-238 is used in breeder reactors, while U-235 is a fissile material widely used in nuclear power and weapons.

Applications of Uranium:

- ❖ Uranium is the primary fuel used in nuclear power plants for electricity generation.
- ❖ It is used in the production of nuclear weapons and military armor, especially depleted uranium (DU).
- ❖ Uranium isotopes are used in radiometric dating, especially U-Pb dating, to determine the age of rocks and Earth's crust.

3. Protactinium (Pa)

Protactinium (atomic number 91) is a rare and highly radioactive element. It is found in very small quantities in uranium ores as a decay product of uranium-235. Due to its scarcity and strong radioactivity, it has no significant industrial applications but is of academic interest.

Applications of Protactinium:

- ❖ Protactinium is used mainly in scientific and nuclear research to study radioactive decay chains and actinide behavior.

4. Actinium (Ac)

Actinium (atomic number 89) is present in trace amounts in uranium ores. It is a soft, silvery radioactive metal and forms as a decay product of uranium. Though rare, it is very useful in certain nuclear and medical applications due to its strong alpha radiation.

Applications of Actinium:

- ❖ Actinium-225 is used in targeted alpha therapy (TAT) for the treatment of cancer.
- ❖ Actinium is also used in neutron sources when combined with beryllium (Ac-Be) for research and scientific instruments.

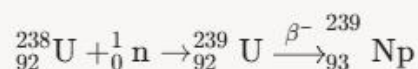
Q. Give some examples of Actinides produced by artificial radioactivity?

[Haq Nawaz bhatti page number 484](#)

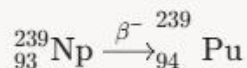
Artificial radioactivity allows scientists to create actinides that do not naturally exist in significant amounts. These transuranic elements (with atomic numbers greater than 92) are mainly produced by bombarding uranium or plutonium with neutrons or alpha particles in nuclear reactors or particle accelerators.

Actinides produced by artificial radioactivity are typically transuranic elements synthesized through nuclear reactions like neutron capture or heavy ion bombardment. Examples include:

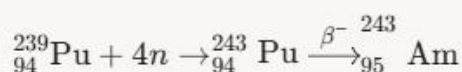
- **Neptunium (Np, Z=93):** Formed by neutron capture on uranium-238, followed by beta decay:



- **Plutonium (Pu, Z=94):** Produced from neptunium-239 via beta decay:



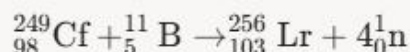
- **Americium (Am, Z=95):** Synthesized by neutron capture on plutonium-239, followed by beta decay:



- **Curium (Cm, Z=96):** Formed by neutron capture on americium-243, followed by beta decay:



- **Lawrencium (Lr, Z=103):** Produced via heavy ion bombardment, for example:



Q. Which of the complex ion namely $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is an outer orbital complex and why?

$[\text{Fe}(\text{CN})_6]^{3-}$ – Inner Orbital Complex

In this complex, iron is in the +3 oxidation state (Fe^{3+}), which gives it a $3d^5$ electronic configuration. The ligand CN^- is a strong field ligand, which causes the pairing of electrons in the 3d orbitals. Due to this pairing, two 3d orbitals become vacant and are available for bonding. These two inner 3d orbitals, along with one 4s and three 4p orbitals, participate in hybridization to form d^2sp^3 hybridization. Since the complex uses inner $(n-1)d$ orbitals, it is classified as an inner orbital complex. Because of electron pairing, the complex is low-spin, and it contains only one unpaired electron, making it weakly paramagnetic.

$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ – Outer Orbital Complex

In this complex, iron is also in the +3 oxidation state (Fe^{3+}) with a $3d^5$ configuration. The ligand H_2O is a weak field ligand, which does not cause pairing of the 3d electrons. As a result, all five 3d orbitals remain singly occupied and are not available for bonding. Therefore, the complex uses outer orbitals—one 4s, three 4p, and two 4d orbitals—for bonding, resulting in sp^3d^2 hybridization. Since outer $(n)d$ orbitals are involved, it is called an outer orbital complex. Because no electron pairing occurs, the complex is high-spin with five unpaired electrons, making it strongly paramagnetic.

[Structure and electronic configuration available sanullah book page number 205 and 205](#)

Q. What is the difference in optical and geometrical isomerism?

Q. Give two applications of Lanthanide alloys?

Haq Nawaz bhatti page number 465

1. Permanent Magnets (NdFeB Alloys)

Lanthanide alloys play a crucial role in modern technology, particularly in the production of high-strength neodymium-iron-boron (NdFeB) magnets. These magnets are the strongest permanent magnets available, offering exceptional magnetic properties that make them indispensable in various applications. They are widely used in electric vehicle motors, where they improve efficiency and power output, and in wind turbine generators, enabling renewable energy production. Additionally, NdFeB magnets are found in everyday electronics like headphones, hard disk drives, and MRI machines, showcasing their versatility and importance in both industrial and consumer sectors.

2. Pyrophoric and Structural Alloys (Mischmetal)

Another significant application of lanthanide alloys is mischmetal, a blend of cerium, lanthanum, and other light lanthanides with iron. This alloy is highly valued for its pyrophoric properties, making it ideal for flints in lighters and ignition devices, where it produces sparks upon friction. Beyond pyrophoric uses, mischmetal is added to magnesium and steel alloys to enhance their strength, durability, and workability. For instance, it is used in military applications like bullets and shells, as well as in aerospace and automotive components, where lightweight yet robust materials are essential.

Q. How MOT is superior to VBT?

VBT	MOT
1. According to VB approach only valence electrons are involved in bond formation.	1. According to MO approach all the electrons of interacting atoms are involved in bond formation.
2. In VBT, the two concerned atoms do not lose their individual identity.	2. In MOT, the two atoms lose their individual identity.
3. It does not offer a simple method of excited states in molecules.	3. It offers a simple and more convenient description of excited states of molecules.
4. Resonance plays an important role in VBT.	4. Resonance plays no role in MOT.
5. It does not explain the paramagnetic behaviour of molecules like O_2 molecule.	5. It explains the paramagnetic behaviour of molecules like O_2 molecule.

Haq Nawaz bhatti page number 194

Q. How magnetism is explained by CFT?

[Haq Nawaz bhatti page number 279](#)

221

Chemistry of d-Block Elements

Equation (iv) shows that the magnitude of μ depends on the number of unpaired electrons (n) present in a given species. The value of μ for different number of unpaired electrons can be calculated with the help of equation (iv) as shown below:

For $n = 0$ $\mu_s = \sqrt{0(0+2)} = 0.0 \text{ B.M.}$

For $n = 1$ $\mu_s = \sqrt{1(1+2)} = \sqrt{3} = 1.73 \text{ B.M.}$

5.7.9 Crystal Field Theory (CFT) and Magnetic Properties of Complexes:

CFT has helped us to understand the magnetic properties of coordination compounds. The substances which have all paired electrons are called diamagnetic. The substances which contain unpaired electrons are called paramagnetic substances. Magnitude of Δ_0 and pairing energy P play an important role in this direction.

- If $\Delta_0 > P$, the electrons will prefer to pair up and the complex will be **low spin complex**.
- If $P > \Delta_0$, the electrons will not pair up and the complex will be **high spin complex**.

Thus, the complexes with **weak field ligands** are **high spin complexes** and are paramagnetic. Those with **strong field ligand** are **low spin complexes** and are diamagnetic.

This generalization could not be provided by valence bond theory. Consider the following two complexes of cobalt (III). From experiments, it has been observed that the complex $[\text{Co}(\text{NH}_3)_6]^{3+}$ is diamagnetic while the complex $[\text{CoF}_6]^{3-}$ is paramagnetic.

Cobalt in $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{CoF}_6]^{3-}$ is in +3 oxidation state and it has d^6 configuration (Fig. (5.32)). It has been observed that F^- ion is a weak field ligand and therefore Δ_0 is small. Therefore, Δ_0 will be less than P and the electron will remain unpaired as far as possible. This is obviously (صاف اندازے) because greater energy is required in pairing and smaller energy is required in moving to orbital of higher energy.

$\begin{array}{c} \text{High spin complex} \\ (\text{Paramagnetic}) \end{array}$
 $\begin{array}{c} \text{Low spin complex} \\ (\text{Diamagnetic}) \end{array}$

Fig. (5.32): Crystal field splitting and explanation for $[\text{CoF}_6]^{3-}$ and $[\text{Co}(\text{NH}_3)_6]^{3+}$ complexes.

R. What is the shape of AB_3E_2 molecule according to VSEPR theory?

9. AB_3E_2 Type

Here A is the central atom, B are the bonded atoms to the central atom and E are the lone pairs of electrons around the central atom. Species of this type have three bonding electron pair and two lone pairs of electrons ($bp = 3$, $lp = 2$). These bonding electron pairs and lone pairs of electrons occupy their positions as far apart as possible so that the electrostatic repulsions between them is minimum. This is most probable if the species of this type has T-shaped bent geometry in which both the lone pairs occupy the equatorial positions of the trigonal bipyramidal geometry. This position of lone pairs involves minimum repulsions as compared to axial position. The bond angle between two axial B-atoms is less than 180° and between equatorial and axial atoms is less than 90° . Example, ClF_3 , BrF_3 , IF_3 , ICl_3 etc. All these molecules have T-shaped geometry.

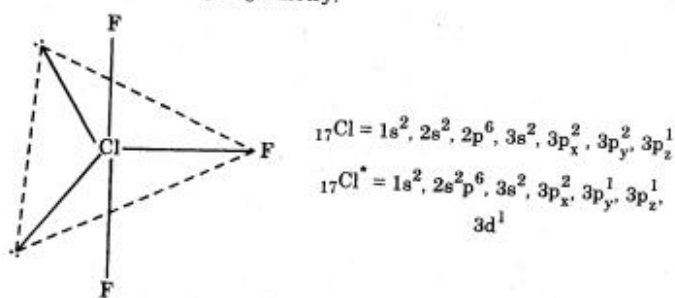


Fig.3.9: T-shaped geometry of ClF_3 molecule

Q.What is the formula to calculate stability constant?

The stability constant quantifies the equilibrium between a metal ion (M) and ligands (L) forming a complex ion (ML_n) in solution:

$$K_{stab} = \frac{[ML_n]}{[M][L]^n}$$

Where:

- $[ML_n]$ = Equilibrium concentration of the complex ion.
- $[M]$ = Equilibrium concentration of the free metal ion.
- $[L]$ = Equilibrium concentration of the free ligand.
- n = Number of ligands bound to the metal ion.

Example: $[Cu(NH_3)_4]^{2+}$ Formation

For the reaction:



The stability constant is:

$$K_{stab} = \frac{[Cu(NH_3)_4(H_2O)_2]^{2+}}{[Cu(H_2O)_6]^{2+}[NH_3]^4}$$

Long Question

Q. Explain how the MOT is used to predict the spectroscopic and magnetic properties transition metal coordination compounds?

[Ansawr on last page of this pdf ,](#)

[Read firstly CFT spectroscopic and magnetic properties Available on haq Nawaz bhatti book , then compare both Mot and CFT because you better understand these topics and very important topics exam point of view.](#)

Q.How uranium is extracted from its ores?

[Haq nawaz bhatti page 482 with flow sheet diagram](#)

Q.Discuss the uses of metal complexes in chemistry and industry?

[Haq nawaz bhatti page 373 Onward](#)

**Beautiful Dua
from Quran**

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

Fall (2023)



Q.What are the structural theories of coordination complexes?

The structural theories of coordination complexes explain the bonding, geometry, and properties of these compounds. The main theories are:

1. Werner's Theory

Werner's Theory of Coordination Compounds defines coordination compounds as entities where the central metal atom exhibits two types of valencies: primary valency and secondary valency. The primary valency corresponds to the oxidation state of the metal and is ionizable, satisfied by negative ions, and written outside the coordination sphere. The secondary valency corresponds to the coordination number, is non-ionizable, satisfied by ligands (neutral molecules or negative ions), and written inside the coordination sphere. Werner's theory explains that the spatial arrangement of ligands around the metal ion is fixed and gives the complex a definite geometry, such as octahedral or tetrahedral. This theory was the first to successfully explain the structures and isomerism of

coordination compounds by distinguishing between these two valencies, although it could not explain properties like color or magnetism

2. Valence Bond Theory (VBT)

Valence Bond Theory (VBT) defines the bonding in coordination complexes as the formation of coordinate covalent bonds through the overlap of ligand lone pair orbitals with vacant hybrid orbitals of the central metal ion. According to VBT, the metal ion undergoes hybridization (**such as d^2sp^3 , sp^3d , or sp^3**) to accommodate ligands, which determines the geometry of the complex. This theory explains the magnetic properties and geometries of complexes by considering the involvement of inner or outer d orbitals, distinguishing between low-spin and high-spin complexes. VBT provides a qualitative understanding of bonding and geometry but does not fully account for electronic spectra or the nature of metal-ligand bonding beyond covalent interactions.

3. Crystal Field Theory (CFT)

Crystal Field Theory (CFT) defines the interaction between the metal ion and ligands as purely electrostatic, where ligands are treated as point charges that cause the splitting of the metal ion's d orbitals into groups of different energies. This splitting explains the color, magnetic behavior, and stability of coordination complexes. CFT predicts the geometry of complexes (octahedral, tetrahedral, square planar) based on the pattern of d orbital splitting and the ligand field strength. It provides a quantitative approach to understanding electronic transitions and magnetic properties but neglects covalent bonding aspects.

4. Ligand Field Theory (LFT) / Molecular Orbital Theory (MOT)

Ligand Field Theory (LFT), an extension of CFT, defines the bonding in coordination complexes by incorporating molecular orbital theory to describe the covalent character of metal-ligand bonds. LFT considers the mixing of metal d orbitals with ligand orbitals, providing a more detailed and accurate description of electronic structure, bonding, and magnetic properties. It explains phenomena that CFT and VBT cannot fully address, such as the nature of bonding orbitals and the influence of ligand symmetry on complex properties.

Q.What is hybridization in AB₆ molecules?

6. sp^3d^2 (or d^2sp^3) Hybridization

One s, three p and two d orbitals may hybridize to give six hybrid orbitals directed at the corners of a regular octahedron. The hybrid orbitals are all equivalent with $16\frac{2}{3}\%$ s, 50% p and $33\frac{1}{3}\%$ d character. SF_6 , IF_5 and ICl_4^- involve this type of hybrid orbitals. In the last two one or two positions respectively are occupied by lone pairs.

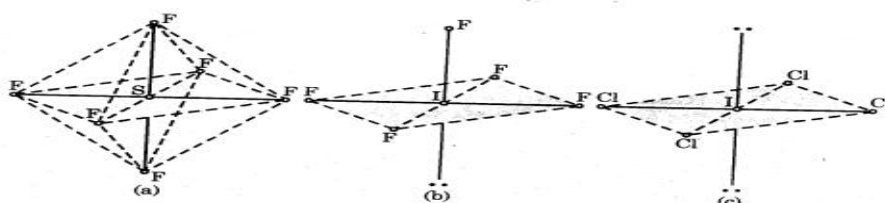
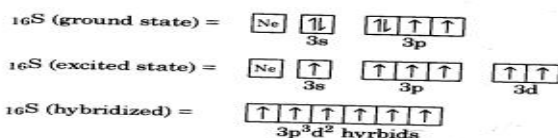


Fig. 3.26: Structures of SF_6 , IF_5 and ICl_4^-

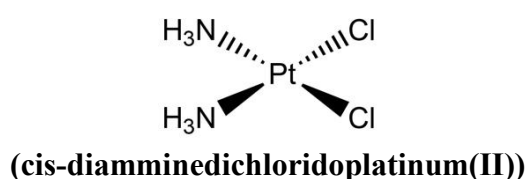
Q.Which theory explains paramagnetism of oxygen?

The theory that successfully explains the paramagnetism of **oxygen (O₂)** is **Molecular Orbital Theory (MOT)**.

While Valence Bond Theory (VBT) and Lewis structures fail to account for oxygen's observed paramagnetism (they predict all electrons to be paired, making it diamagnetic), MOT accurately predicts that the oxygen molecule has two unpaired electrons in its antibonding **π^* (pi-star) ($\pi^*_{2p_y}$ and $\pi^*_{2p_z}$)** molecular orbitals. These unpaired electrons give the oxygen molecule a net magnetic moment, causing it to be attracted to a magnetic field, thus exhibiting paramagnetism.

Structure for VBT and MOT Repeat

Q. Which metal complex is used in cancer therapy?



The most prominent and widely used metal complexes in cancer therapy are platinum-based drugs, with cisplatin (**cis-diamminedichloridoplatinum(II)**) [**Pt(NH₃)₂Cl₂**] being the pioneering and most famous example. Discovered in the late **1960s** and approved by the FDA in **1978**, cisplatin is highly effective against various solid tumors, including ovarian, testicular, bladder, lung, head and neck cancers, and some lymphomas. Its anticancer activity arises from forming DNA cross-links that disrupt replication and transcription, leading to cancer cell death (apoptosis).

Due to cisplatin's significant side effects such as nephrotoxicity, neurotoxicity, and ototoxicity, other platinum drugs like **carboplatin** and **oxaliplatin** were developed. Carboplatin has a different toxicity profile with less kidney damage but more bone marrow suppression, while oxaliplatin is mainly used for colorectal cancer and causes peripheral neuropathy.

Q. Why are f-block elements are called rare earth elements?

The f-block elements, particularly the lanthanides (**atomic numbers 57-71**), are commonly referred to as rare earth elements due to a combination of historical and geological factors. When these elements were first discovered in the late **18th and 19th centuries**, they were initially thought to be extremely scarce in nature, hence the term "**rare**." However, we now know that many of these elements are actually relatively abundant in Earth's crust - some, like cerium, are nearly as common as copper. The "earth" part of their name comes from early chemists who classified metal oxides as "**earths**," as these elements were typically isolated from mineral oxides like **yttria and ceria**.

These elements are called "**rare**" because they are spread out in small amounts in many different minerals, not found in large, concentrated deposits. Also, they are very similar to each other chemically, which makes it hard and expensive to separate them from each other. Because of these difficulties, they kept the name "rare earth elements" even after scientists realized they are not actually very rare. Today, these elements are very important for many modern technologies like smartphones, electric cars, wind turbines, and military equipment.

Haq Nawaz Bhatti Page Number 440

Q.How are coordination complexes prepared?

[Haq Nawaz Bhatti Page Number 310 to 314 , you can write according your need](#)

Q.How lanthanides can be separated from actinides?

[Repeat in Fall 2024](#)

Q. What is the type of hybridization in $\text{Fe}(\text{CO})_5$ and $[\text{Ni}(\text{CN})_4]^{2-}$?

4. dsp^2 Hybridization

dsp^2 hybridization involves the mixing of one s, two p and one d orbitals on the same atom to form four identical dsp^2 hybrids. Each four hybrid orbitals are directed to the corners of a square giving a square planar structure. The formation of $[\text{Ni}(\text{CN})_4]^{2-}$ ion can be explained on dsp^2 hybridization. This molecule has square planar geometry as shown in figure 3.22. This hybridization is mostly encountered in coordination compounds.

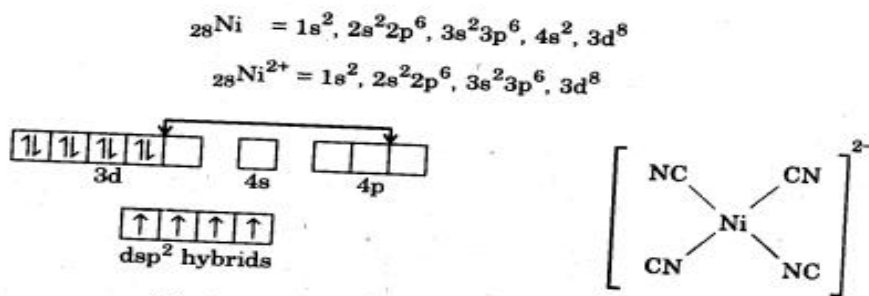


Fig.3.22: Formation of $[\text{Ni}(\text{CN})_4]^{2-}$ ion.

$\text{Fe}(\text{CO})_5$

[Available Sanaullah book page number 203 onward , Same concept above mention](#)

(a) How does the coordination number of a metal affect the stability of a complex?

[Haq Nawaz Bhatti Page Number 306 to 310, you can write according your need](#)

Q.How are the colors of coordination compounds explained by CFT?

[Haq Nawaz Bhatti Page Number 276](#)

OR

The colors of coordination compounds are explained by Crystal Field Theory through the splitting of the metal ion's d-orbitals in the presence of ligands and the resulting electronic transitions known

as d-d transitions. When ligands approach a metal ion, the five degenerate d-orbitals split into different energy levels depending on the geometry of the complex. **For example**, in an octahedral complex, the d-orbitals split into a lower-energy set called **t_{2g}** and a higher-energy set called **e_g**, separated by an energy gap called the crystal field splitting energy Δ_0 .

When white light shines on the complex, electrons in the lower-energy t_{2g} orbitals absorb light energy equal to Δ_0 and get excited to the higher-energy e_g orbitals. The wavelength of light absorbed corresponds to this energy difference. The color we observe is the complementary color of the absorbed light because the unabsorbed wavelengths are transmitted or reflected.

Several factors influence the color of coordination compounds:

Oxidation state of the metal:

Higher oxidation states increase Δ_0 , shifting absorption to shorter wavelengths (blue shift).

Nature of ligands:

Strong-field ligands (**like CN⁻ or CO**) cause larger splitting and absorption of higher-energy light, while weak-field ligands (**like H₂O or Cl⁻**) cause smaller splitting.

Coordination geometry:

Different geometries (octahedral, tetrahedral, square planar) produce different splitting patterns and magnitudes, affecting color.

Some complexes with **d⁰ or d¹⁰** configurations do not show color because they lack d-d transitions. Additionally, some colors arise from charge transfer transitions rather than d-d transitions.

Example:

The complex [Ti(H₂O)₆]³⁺ absorbs green light (~500 nm), so it appears purple, the complementary color of green.

CFT explains the color of coordination compounds by the splitting of d-orbitals in the ligand field, absorption of specific wavelengths of light corresponding to this splitting, and the transmission of the complementary color.

Q.What is the difference between a hard and a soft ligand in coordination chemistry?

Sanaullah Page Number 138 to 139 Answer in symbiosis paragraph

OR

In coordination chemistry, ligands are classified as hard or soft based on their electronic and structural properties, which influence their bonding preferences with metal ions.

Hard ligands

Hard ligands are typically small, have high electronegativity, and low polarizability. They usually contain donor atoms like oxygen, nitrogen, or fluorine that hold their electrons tightly. Hard ligands tend to form ionic or electrostatic bonds with metal ions, preferring to bind to hard metal ions (which are small with high positive charge).

Examples of hard ligands include F⁻, OH⁻, NH₃, and H₂O.

Soft ligands

Soft ligands, on the other hand, are larger, more polarizable, and have lower electronegativity. They contain donor atoms such as sulfur, phosphorus, or iodine, with electrons more easily distorted. Soft ligands tend to form covalent bonds with metal ions and prefer to bind to soft metal ions (which are larger with lower positive charge and often have filled or partially filled d orbitals).

Examples of soft ligands include I^- , S^{2-} , PR_3 (phosphines), and CN^- .

This classification is part of the Hard and Soft Acids and Bases (HSAB) theory, which states that hard acids prefer to bind to hard bases (hard ligands), and soft acids prefer to bind to soft bases (soft ligands), resulting in more stable complexes.

Q.What is the difference between square planar and tetrahedral complexes?

[https://chem.libretexts.org/Courses/Douglas_College/DC%3A_Chem_2330_\(O'Connor\)/4%3A_Crystal_Field_Theory/4.3%3A_High_Spin_and_Low_Spin_Complexes](https://chem.libretexts.org/Courses/Douglas_College/DC%3A_Chem_2330_(O'Connor)/4%3A_Crystal_Field_Theory/4.3%3A_High_Spin_and_Low_Spin_Complexes)

OR

<https://www.differencebetween.com/difference-between-square-planar-and-tetrahedral-complexes/>

Aspect	Square Planar Complexes	Tetrahedral Complexes
Geometry	The four ligands are arranged at the corners of a square in the same plane around the central metal ion.	The four ligands are positioned at the corners of a tetrahedron around the central metal ion.
Bond Angles	The bond angles between adjacent ligands are 90° , and trans ligands are 180° apart.	The bond angles between all ligands are approximately 109.5° .
Coordination Number	The coordination number is 4.	The coordination number is also 4.
Common Metal Electron Config.	Most common for d^8 metals (e.g., Ni^{2+} , Pd^{2+} , Pt^{2+} , Au^{3+}) with strong-field ligands .	Common for d^0 , d^{10} , or weak-field d^7/d^9 metals (e.g., Zn^{2+} , Cd^{2+} , Co^{2+} , Cu^+).
Crystal Field Splitting	The d-orbitals split into four levels : $d_{x^2-y^2}$ (highest energy) $> d_{z^2} > d_{xy} > d_{xz}/d_{yz}$.	The d-orbitals split into two levels : t_2 (d_{xy} , d_{xz} , d_{yz}) $> e$ (d_{z^2} , $d_{x^2-y^2}$). Splitting energy (Δ) is smaller ($\approx 4/9 \Delta_o$).
Magnetic Properties	Usually low-spin/diamagnetic (e.g., $[Ni(CN)_4]^{2-}$).	Often high-spin/paramagnetic (e.g., $[CoCl_4]^{2-}$), except d^{10} (e.g., $[ZnCl_4]^{2-}$).
Examples	$[Pt(NH_3)_2Cl_2]$ (cisplatin), $[Ni(CN)_4]^{2-}$, $[PdCl_4]^{2-}$.	$[ZnCl_4]^{2-}$, $[MnCl_4]^{2-}$, $[NiCl_4]^{2-}$ (high-spin Ni^{2+}).

Q.How the stability of a coordination compound can be checked?

[Repeat](#)

Q.How do coordination compounds play a role in biological systems?

[Repeat in Fall 2024](#)

Q.What is the difference between high spin and low spin complex in coordination chemistry?

[Haq Nawaz Bhatti Page Number 270 Onward](#)

Aspect	High Spin Complex	Low Spin Complex
Ligand Type	High spin complexes are formed when the ligands are weak field ligands, such as water (H_2O), fluoride (F^-), or chloride (Cl^-).	Low spin complexes are formed when the ligands are strong field ligands, such as cyanide (CN^-), carbonyl (CO), or ammonia (NH_3).
Crystal Field Splitting	In high spin complexes, the crystal field splitting energy (Δ) is small, so electrons can occupy higher orbitals easily.	In low spin complexes, the crystal field splitting energy (Δ) is large, so electrons prefer to pair up in lower orbitals.
Electron Configuration	Electrons tend to occupy all orbitals singly first before pairing (Hund's rule).	Due to large splitting, electrons in low spin complexes prefer to pair in the lower energy orbitals to avoid high energy states.
Number of Unpaired Electrons	High spin complexes usually have a greater number of unpaired electrons, depending on the metal and its oxidation state.	Low spin complexes usually have fewer or no unpaired electrons, leading to different magnetic properties.
Magnetic Properties	High spin complexes are generally paramagnetic because of the presence of unpaired electrons.	Low spin complexes are often diamagnetic or weakly paramagnetic, due to electron pairing.
Examples	An example of a high spin complex is $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, where water is a weak field ligand and more electrons remain unpaired.	An example of a low spin complex is $[\text{Fe}(\text{CN})_6]^{3-}$, where cyanide is a strong field ligand and electrons pair up in lower orbitals.

Long Questions

Q.How uraniium is extracted from its ores with flow sheet daigram ?

[Repeat](#)

**Q. What is the importance of actinides and lanthanides in chemistry and industry?
How they differ from each other?**

[Haq nawaz Bhaati Page number 477 onward and applications available](#)

Q. Explains the types of isomerisms with suitable examples ?

[Repeat](#)

Beautiful Dua from Quran

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

Spring (2023)



Q.What is the electronic configuration of the Lanthanides and Actinides?

7.4 ELECTRONIC CONFIGURATIONS OF LANTHANIDES

It may be noted that atoms of these elements have electronic configuration with $6s^2$ common but with variable occupancy of $4f$ level (Table 7.2). However, the electronic configurations of all the tripositive ions (the most stable oxidation state of all the lanthanides) are of the form $4f^n$ ($n = 1$ to 14 with increasing atomic number).

Table 7.2: Electronic configurations

Element	Ln	Ln ²⁺	Ln ³⁺
La	5d ¹ 6s ²	5d ¹	4f ⁰
Ce	4f ¹ 5d ¹ 6s ²	4f ²	4f ¹
Pr	4f ³ 6s ²	4f ³	4f ²
Nd	4f ⁴ 6s ²	4f ⁴	4f ³
Pm	4f ⁵ 6s ²	4f ⁵	4f ⁴
Sm	4f ⁶ 6s ²	4f ⁶	4f ⁵
Eu	4f ⁷ 6f ²	4f ⁷	4f ⁶
Gd	4f ⁷ 5d ¹ 6s ²	4f ⁷ 5d ¹	4f ⁷
Tb	4f ⁹ 6f ²	4f ⁹	4f ⁸
Dy	4f ¹⁰ 6s ²	4f ¹⁰	4f ⁹
Ho	4f ¹¹ 6s ²	4f ¹¹	4f ¹⁰

7.14 ELECTRONIC CONFIGURATIONS OF ACTINIDES AND IONIC SIZES.

All the actinides are believed to have the electronic configuration of $7s^2$ and variable occupancy of the $5f$ and $6d$ subshells. The fourteen electrons are formally added to $5f$, though not in thorium ($Z = 90$) but from Pa onwards the $5f$ orbitals are complete at element 103.

The irregularities in the electronic configurations of the actinoids, like those in the lanthanoids are related to the stabilities of the f^0 , f^7 and f^{14} occupancies of the $5f$ orbitals. Thus, the configurations of Am and Cm are $[Rn] 5f^7 7s^2$ and $[Rn] 5f^7 6d^1 7s^2$.

Table 7.9: Electronic Configurations of Actinides

89.	Actinium	(Ac)	$[Rn] 5f^0, 6d^1, 7s^2$
90.	Thorium	(Th)	$[Rn] 5f^0, 6d^2, 7s^2$
91.	Protoactinium	(Pa)	$[Rn] 5f^2, 6d^1, 7s^2$
92.	Uranium	(U)	$[Rn] 5f^3, 6d^1, 7s^2$
93.	Neptunium	(Np)	$[Rn] 5f^4, 6d^1, 7s^2$
94.	Plutonium	(Pu)	$[Rn] 5f^6, 6d^0, 7s^2$
95.	Americium	(Am)	$[Rn] 5f^7, 6d^0, 7s^2$
96.	Curium	(Cm)	$[Rn] 5f^7, 6d^1, 7s^2$
97.	Berkelium	(Bk)	$[Rn] 5f^9, 6d^0, 7s^2$
98.	Californium	(Cf)	$[Rn] 5f^{10}, 6d^0, 7s^2$
99.	Einsteinium	(Es)	$[Rn] 5f^{11}, 6d^0, 7s^2$
100.	Fermium	(Fm)	$[Rn] 5f^{12}, 6d^0, 7s^2$
101.	Mendelevium	(Md)	$[Rn] 5f^{13}, 6d^0, 7s^2$
102.	Nobelium	(No)	$[Rn] 5f^{14}, 6d^0, 7s^2$
103.	Lawrencium	(Lr)	$[Rn] 5f^{14}, 6d^1, 7s^2$

Q. Why are the Lanthanides and Actinides also known as Rare Earth elements?

Lanthanides and actinides are often called "rare earth elements" due to historical, mineralogical, and chemical reasons:

❖ Historical Origin of the Term "Rare Earth":

In the 18th and 19th centuries, lanthanides were discovered in uncommon mineral ores like monazite, gadolinite, and xenotime. These elements were found as oxides (then called "earths") and were difficult to isolate. Early chemists believed they were rare in nature, so the name "rare earths" was used—even though many are now known to be quite abundant.

❖ Chemical Similarity and Extraction Difficulty:

Lanthanides occur together in ores and have very similar chemical properties, especially a common +3 oxidation state. This made their separation extremely difficult using early techniques. Even today, advanced methods like solvent extraction and ion exchange are needed to isolate them.

❖ Abundance Misconception:

Although the name suggests rarity, many rare earth elements (like cerium, neodymium, and lanthanum) are more abundant than lead or copper in Earth's crust. The term "rare" reflects the difficulty in finding concentrated deposits, not their actual quantity.

❖ Actinides and Their Classification:

Actinides ($5f$ elements) share chemical similarities with lanthanides but are mostly radioactive

and found in different ways. Only thorium and uranium occur naturally in significant amounts. Because of their radioactivity and nuclear applications, actinides are not typically included in the rare earth category—though they are related.

❖ **Scandium and Yttrium Inclusion:**

Scandium and yttrium are also included with rare earths because they are found in the same ores and have similar chemical properties, though not part of the lanthanide series.

Q. Where are the Lanthanides and Actinides found in the periodic table?

Lanthanides

Lanthanides are a group of 15 metallic elements from atomic number 57 (Lanthanum, La) to 71 (Lutetium, Lu). These elements are known as 4f-block elements because their electrons progressively fill the 4f orbital. They are also referred to as rare earth metals and are considered inner transition elements due to their electron configuration.

Position in Periodic Table:

- ❖ Found in the 6th period.
- ❖ Belong to the f-block.
- ❖ Placed below the main body of the periodic table for convenience.
- ❖ Technically positioned between Lanthanum (La) and Hafnium (Hf) in Group 3.

List of Lanthanides (Atomic No. 57–71):

La (Lanthanum), Ce (Cerium), Pr (Praseodymium), Nd (Neodymium), Pm (Promethium), Sm (Samarium), Eu (Europium), Gd (Gadolinium), Tb (Terbium), Dy (Dysprosium), Ho (Holmium), Er (Erbium), Tm (Thulium), Yb (Ytterbium), Lu (Lutetium)

Actinides

Actinides are a series of 15 elements from atomic number 89 (Actinium, Ac) to 103 (Lawrencium, Lr). These are 5f-block elements, meaning their electrons fill the 5f orbitals. Like lanthanides, actinides are also inner transition metals, and many of them are radioactive, especially the heavier ones.

Position in Periodic Table:

- ❖ Located in the 7th period.
- ❖ Belong to the f-block.
- ❖ Displayed below the lanthanides in a separate row.
- ❖ Positioned between Actinium (Ac) and Rutherfordium (Rf) in Group 3.

List of Actinides (Atomic No. 89–103):

Ac (Actinium), Th (Thorium), Pa (Protactinium), U (Uranium), Np (Neptunium), Pu (Plutonium), Am (Americium), Cm (Curium), Bk (Berkelium), Cf (Californium), Es (Einsteinium), Fm (Fermium), Md (Mendelevium), No (Nobelium), Lr (Lawrencium)

Q.Differentiate between inner and outer orbital complexes?

Repeat

Q.Give limitations of VBT?

Limitations of VBT

The valence bond theory was fairly successful in explaining qualitatively the geometry of complexes, but still it has following limitations:

1. The valence bond theory cannot account for the relative stabilities for different shapes and different coordination numbers in metal complexes.
2. The VBT cannot explain the magnetic moment data of some complexes.
3. This theory does not explain the absorption spectra, colour of transition metal ions and heats of formation etc.
4. This theory is unable to explain the nature of some of the complexes of copper.
5. This theory cannot account for the relative rates of reactions of analogous metal complexes.
6. Too much stress has been laid on the metal ion while the importance of ligand is not properly stressed.
7. It cannot explain that why did different complexes of the same metal show different colours.
8. Relative stabilities of different complexes could not be explained.
9. It cannot explain that why should certain ligands form high spin, while others low spin complexes.

Q.Explain the magnetic behavior complexes by comparing its VBT and MOT picture of iron (II) complexes?

Repeat

Q.Give two applications of metal complexes in biological systems?

Repeat

Q.What is a bent bond?

A bent bond, also known as a **banana bond**, is a type of covalent bond where the bond angle between atoms is less than the normal tetrahedral angle (109.5°), causing the bond to appear curved or "bent" like a banana. This kind of bonding arises when standard orbital hybridization (like sp^3) cannot be maintained due to ring strain or molecular geometry.

Explanation:

- ❖ Bent bonds usually occur in small ring compounds (e.g., cyclopropane) where the ring size is too small to accommodate normal bond angles.
- ❖ In such molecules, rehybridization happens, and orbitals bend to reduce angle strain.
- ❖ The electron density in a bent bond bulges out, forming a curved shape instead of a straight line.
- ❖ These are a special type of sigma bonds where orbitals overlap obliquely.

Example:

Cyclopropane is a classical example:

- ❖ In cyclopropane, the C–C–C bond angle is $\sim 60^\circ$, which is much smaller than the tetrahedral angle (109.5°).
- ❖ Carbon atoms cannot maintain standard sp^3 hybridization, so the bonds become bent to relieve strain.

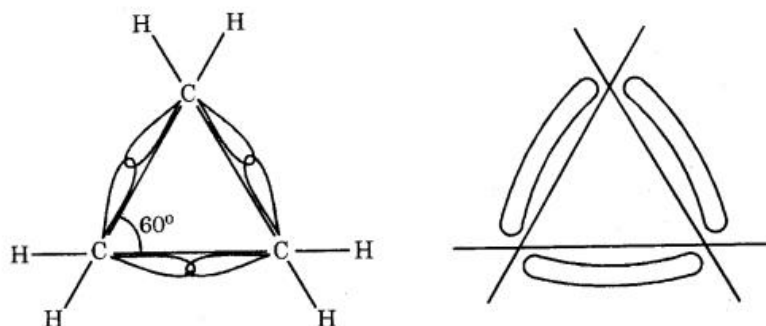


Fig.3.59: Depiction of the banana bond in cyclopropane

As shown in the diagram, the bonds curve outward like a banana, hence the name banana bond.

Q. Give the names of the sub groups of the elements of Lanthanide ores?

Lanthanide ores contain rare earth elements that are commonly classified into the following two main subgroups based on their atomic number, chemical properties, and type of ore deposit:

1. Light Rare Earth Elements (LREEs)

These include lanthanides with lower atomic numbers ($Z = 57$ to 64). They have larger ionic radii and are more abundant in nature.

Common Ores:

- ❖ Monazite
- ❖ Bastnäsite

Elements Included:

Lanthanum (La), Cerium (Ce), Praseodymium (Pr), Neodymium (Nd), Promethium (Pm), Samarium (Sm), Europium (Eu), Gadolinium (Gd)

Applications:

- ❖ Neodymium is used in manufacturing powerful magnets used in wind turbines and electric vehicles.
- ❖ Cerium is used in catalytic converters and for polishing optical glass.
- ❖ Lanthanum is used in camera lenses and hybrid vehicle batteries.

2. Heavy Rare Earth Elements (HREEs)

These include lanthanides with higher atomic numbers ($Z = 65$ to 71) and the element Yttrium (Y). They have smaller ionic radii and are less abundant.

Common Ores:

- ❖ Xenotime
- ❖ Ionic clays

Elements Included:

Terbium (Tb), Dysprosium (Dy), Holmium (Ho), Erbium (Er), Thulium, (Tm) Ytterbium (Yb), Lutetium (Lu), Yttrium (Y) (included due to chemical similarity)

Applications:

- ❖ Terbium is used in green phosphors for LED lighting and electronic displays.
- ❖ Dysprosium is used in heat-resistant magnets for motors and generators.
- ❖ Gadolinium is used in MRI machines as a contrast agent in medical imaging.

Q.How CFT explains magnetic properties of metal complexes?

[Repeat](#)

Q.Explain the structure of $[\text{Ni}(\text{CN})_4]$ according to valence bond theory?

[Repeat](#)

Q.What is the role of spectrochemical series?

1. Determines Crystal Field Splitting (Δ_o):

The spectrochemical series plays a key role in coordination chemistry by arranging ligands based on their ability to cause d-orbital splitting in a metal ion. This energy difference is called crystal field splitting energy (Δ_o). Weak-field ligands like I^- and Br^- cause small splitting, while strong-field ligands like CN^- and CO cause large splitting. The magnitude of this splitting helps determine the behavior of the complex.

2. Predicts Spin States (High Spin vs. Low Spin):

The spectrochemical series helps predict whether a complex will be high-spin or low-spin. In the presence of **weak-field ligands**, the d-orbital splitting is small, so electrons remain unpaired in higher orbitals, forming a high-spin complex. In contrast, **strong-field ligands** create larger splitting, causing electrons to pair up in lower orbitals, resulting in a low-spin complex. This directly affects the magnetic properties of the compound.

3. Explains Color of Complexes:

The spectrochemical series also helps explain the color of coordination compounds. The crystal field **splitting energy** (Δ_o) corresponds to the energy of light absorbed during **d–d electronic transitions**. The wavelength of light absorbed determines the color we observe. Complexes with large Δ_o absorb higher-energy (shorter wavelength) light and show different colors than those with small Δ_o .

4. Predicts Magnetic Properties:

Since the number of unpaired electrons depends on whether the complex is high-spin or low-spin, the spectrochemical series helps predict its magnetic behavior. High-spin complexes usually have more unpaired electrons and are therefore paramagnetic (attracted to a magnetic field), while low-spin complexes have fewer or no unpaired electrons and are often diamagnetic (repelled by a magnetic field).

5. Guides Complex Stability and Reactivity:

The spectrochemical series also influences the stability and reactivity of complexes. Strong-field ligands generally provide greater crystal field stabilization energy (**CFSE**), making the complex more thermodynamically stable. This also affects how easily the complex undergoes substitution reactions, helping chemists understand and predict its chemical behavior.

6. Helps in Complex Design:

Chemists use the spectrochemical series to design metal complexes with specific **colors, magnetic properties, or catalytic activities**. For example, the design of **anticancer** drugs using **platinum complexes or MRI** contrast agents with gadolinium relies on selecting the right ligands from the series. This makes the spectrochemical series a powerful tool in practical and medical applications of coordination chemistry.

Q.What are the uses of Lanthanides and Actinides in industries and technology?

Uses of Lanthanides and Actinides in Industry

Lanthanides are used in industries for making strong magnets, glass polishing powders, and in the manufacturing of alloys and batteries. For example, cerium is used in car catalytic converters and in polishing glass. Lanthanum is used in metal alloys and rechargeable batteries.

Actinides like uranium and plutonium are used in nuclear power plants and submarines as energy sources. Americium is used in smoke detectors in household and industrial safety systems.

Uses of Lanthanides and Actinides in Technology

Lanthanides play a key role in advanced technology such as LED lighting, smartphones, and TV displays—especially europium and terbium which give color to screens. Neodymium is used in computer hard drives and headphones.

In technology, **actinides** such as plutonium-238 are used in spacecraft to generate electricity through radioisotope thermoelectric generators. Some actinides are also used in radiation therapy and scientific instruments involving nuclear research.

Q.Draw the structure of following molecules as per VSEPR model?

AB₇, and AB₄E

AB₇:

Here A is the central atom and B are the atoms bonded to the central atom with no lone pair of electrons (**bp = 7, lp = 0**). These bonding electron pairs occupy positions as far apart as possible to minimize electrostatic repulsions. The most stable geometry for AB₇ molecules is **pentagonal bipyramidal**, where the central atom is surrounded by seven bonding pairs arranged in a pentagonal

plane (**five B atoms**) and two axial positions (**two B atoms**). In this geometry, the **B—A—B** bond angles are **72°** (between adjacent equatorial atoms) and **90°** (between axial and equatorial atoms).

Examples:

- ❖ IF₇ (Iodine heptafluoride)
- ❖ XeF₇⁻ (Unstable, but theoretically possible)

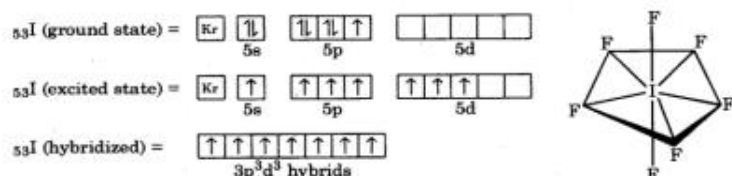


Fig.3.27: Formation of IF₇ molecule having pentagonal bipyramidal geometry

AB₄E:

8. AB₄E Type

Here A is the central atom, B are the bonded atoms to the central atom and E is lone pair of electrons. Species of this type have four bonding electron pairs and one lone pair (bp = 4, lp = 1). These bonding and lone pair of electrons occupy their positions as far apart as possible so that the electrostatic repulsion between them is minimum. This is most probable if the species of this type has distorted tetrahedral

Chemical Bonding 141

geometry in which the lone pair occupies the equatorial position of the trigonal bipyramidal (correct) geometry. In the equatorial position the repulsion between lp and bp is minimum as compared to the repulsion between lp and bp in the axial position. The bond angle between two equatorial B atoms is less than 180°, while the bond angle between two axial B atoms is less than 90°. Examples, SF₄, SeF₄, TeF₄, TeCl₄.

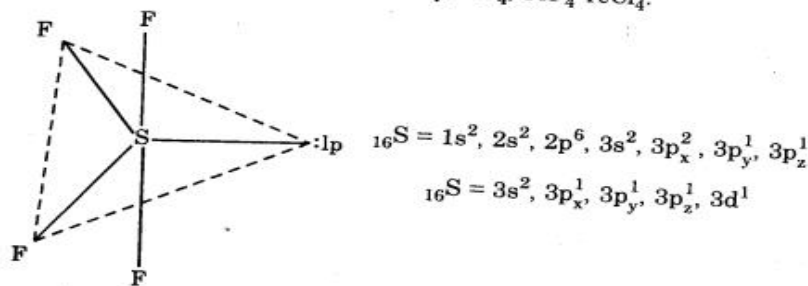


Fig.3.8: Distorted of Geometry of SF₄ molecule

Q.What are geometrical isomers, explain with examples?

Repeat

Long Question

Q.What is Lanthanide contraction? How it effects the properties of Lanthanides?

[Haq Nawaz Bhatti Page number 457 onward](#)

Q.How coordination complexes are prepared. Give details with examples?

[Haq Nawaz Bhatti Page number 310 onward](#)

Q.Explain the geometry of AB₈ and AB₉ type of molecules with reference to hybridization theory?

[Repeat in short questions](#)

**Beautiful Dua
from Quran**

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وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

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Q.What are chelates? Name of factors which influence the stability of chelates.

Chelates:

[Repeat](#)

Factors affecting the stability of chelates

Following are important factors which influence the stability of chelates.

1. **Size of the chelate ring.** Chelates having 6-membered rings including the metal atom are more stable than those having 5-membered rings which in turn, are more stable than the chelates with 4-membered rings and so on.
2. **Number of chelate rings.** Greater in the number of chelate rings, greater is the stability of the chelate.
3. **Resonance effects.** Resonance enhances the stability of the chelate.
4. **Chelate effect.** The chelated complexes are known to be more stable than the non-chelated complexes. This effect is known as chelate effect.

Q.What property of transition metals allows them to form the coordination compound easily?

The main property that enables transition metals to form coordination compounds easily is their capacity to exhibit variable oxidation states, the presence of vacant d-orbitals, small ionic size with high charge density, and the extra stability gained from crystal field stabilization energy (CFSE).

1. Variable Oxidation States

Transition metals have several possible oxidation states because their 3d and 4s electrons are close in energy. This flexibility allows them to combine with many different ligands, forming complexes of varying geometry and reactivity. For example, iron forms **Fe²⁺ in hemoglobin** and **Fe³⁺ in ferric** compounds.

2. Vacant d-Orbitals

Transition metal ions have incomplete d-subshells containing vacant d-orbitals. These orbitals accept lone pairs from ligands, forming coordinate (dative covalent) bonds. For instance, in **[Cu(NH₃)₄]²⁺**, **copper(II)** uses its vacant d-orbitals to bond with ammonia ligands.

3. Small Size and High Charge Density

These ions are small and highly charged, resulting in a strong attraction for ligands via electrostatic forces. As an example, **Cr³⁺** (small size, high charge) forms the stable complex **[Cr(H₂O)₆]³⁺**.

4. Crystal Field Stabilization Energy (CFSE)

When ligands approach, the metal's d-orbitals split into different energy levels (crystal field theory). The arrangement of electrons in these split orbitals increases the stability of the complex due to CFSE. For example, **[Fe(CN)₆]⁴⁻** is highly stabilized due to the strong field ligands (cyanide).

Q.Explain the structure of Hexaammine cobalt (II) chloride and Hexaammineplatinum (IV).

1. Hexaamminecobalt(II) Chloride

Hexaamminecobalt(II) chloride has the chemical formula **[Co(NH₃)₆]Cl₂**. In this complex, the **central metal ion is cobalt (Co²⁺)**, which is in the **+2 oxidation state**. It is surrounded by **six ammonia (NH₃) molecules**, which act as **neutral monodentate ligands**. Each ammonia donates a lone pair of electrons from nitrogen to the cobalt ion, forming **coordinate covalent bonds**. The **coordination number** of cobalt in this complex is **6**, resulting in an **octahedral geometry**, where the six ligands are symmetrically arranged around the metal ion. The overall complex ion formed is **[Co(NH₃)₆]²⁺**, which is balanced by **two chloride ions (Cl⁻)** that are **not directly bonded** to the cobalt but are **present outside the coordination sphere as counter-ions**. These chloride ions help maintain electrical neutrality and are held by **ionic interactions** with the positively charged complex ion. This structure is a classic example of an **octahedral coordination compound** involving a **d⁷ transition metal ion**.

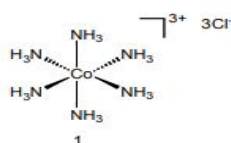
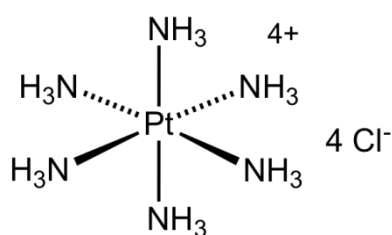


Chart 1: Hexaamminecobalt (II) chloride (Cohex).

[Corrections 2+ , 2Cl⁻](#)

2. Hexaammineplatinum(IV)

Hexaammineplatinum(IV) has the formula $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$. In this coordination compound, the **central metal ion is platinum (Pt^{4+})** in the **+4 oxidation state**. Platinum is coordinated with **six ammonia (NH_3) ligands**, which are **neutral and monodentate**, each donating a lone pair of electrons from nitrogen to the platinum ion to form **coordinate bonds**. The **coordination number** of platinum is **6**, giving rise to an **octahedral geometry** with the six ammonia ligands symmetrically arranged around the central (Pt^{4+}) ion. The complex ion formed is $[\text{Pt}(\text{NH}_3)_6]^{4+}$, and its charge is balanced by **four chloride anions (Cl^-)**, which are **outside the coordination sphere** and held via **ionic interactions**. These chloride ions do not participate in bonding with the platinum directly but ensure that the compound remains electrically neutral. This complex is also a good example of a **stable octahedral geometry**, formed by a **d^6 transition metal ion** in a higher oxidation state.



OR

[This answer according to Warner Theory Haq Nawaz Bhatti page number 233 onward](#)

1. Hexaamminecobalt(II) chloride $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$

This ammine has six NH_3 molecules. All ammonia ligands are neutral and satisfy the secondary valency (coordination number) of cobalt. Cobalt is in the +2 oxidation state. As all six ligands are neutral NH_3 , they do not contribute to the charge of the complex. The cobalt ion thus forms a stable complex ion $[\text{Co}(\text{NH}_3)_6]^{2+}$. The two chloride ions (Cl^-) are ionizable and satisfy the primary valency (oxidation state). These two Cl^- ions lie outside the coordination sphere and can be precipitated as AgCl when treated with AgNO_3 .

2. Hexaammineplatinum(IV) – $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$

This ammine has six NH_3 molecules, all of which are neutral ligands and satisfy the secondary valency (coordination number) of platinum. Platinum is in the +4 oxidation state. The six NH_3 ligands form coordinate bonds with Pt^{4+} , giving the complex ion $[\text{Pt}(\text{NH}_3)_6]^{4+}$. To satisfy the primary valency of +4, four chloride ions (Cl^-) are present outside the coordination sphere. These Cl^- ions are ionizable and can be detected by precipitation with AgNO_3 solution as AgCl .

Q.What is the oxidation states commonly found in Actinides?

[Haq Nawaz Bhatti page number 471 to 473 onward](#)

The actinides exhibit a wide range of oxidation states due to the comparable energies of their 5f, 6d, and 7s orbitals. The most common oxidation state is +3, but higher states (+4 to +7) are also observed, especially in the first half of the series. The distribution of oxidation states is uneven across the actinides, with earlier elements showing more variability than later ones.

1. +2 Oxidation State

The +2 oxidation state is observed in americium (Am) and heavier actinides (Cf, Es, Fm, Md, No). Americium does not form Am^{2+} in aqueous solutions, but solid compounds like AmI_2 exist. Nobelium (No) shows exceptional stability in the +2 state due to its $[\text{Rn}]5f^4 7s^2$ configuration, which results in a fully filled 5f subshell. In fact, No^{2+} is more stable than its lanthanide counterpart Yb^{3+} .

2. +3 Oxidation State

The +3 oxidation state is the most common and stable for most actinides, particularly from americium (Am) onwards. For the first four elements (Th to Np), the +3 state is not the most stable, but it becomes increasingly dominant in the latter half of the series (Am to Lr). The chemistry of these later actinides resembles that of the lanthanides, where the +3 state is predominant.

3. +4 Oxidation State

The +4 oxidation state is exhibited by actinides from thorium (Th) to berkelium (Bk). Thorium exclusively shows the +4 state, forming compounds like ThO_2 and ThCl_4 . For protactinium (Pa), uranium (U), neptunium (Np), plutonium (Pu), and berkelium (Bk), the +4 state is stable in solutions and solid compounds. However, for americium (Am) and curium (Cm), the +4 state is less common and exists mainly as complex fluoro anions.

4. +5 Oxidation State

The +5 oxidation state is observed from protactinium (Pa) to americium (Am). Protactinium shows the greatest stability in this state, resembling niobium (Nb) and tantalum (Ta). Uranium, neptunium, plutonium, and americium also form +5 compounds, such as pentafluorides (e.g., UF_5 , PaF_5). However, these compounds are generally not stable in aqueous solutions, where oxo-cations like MO^{2+} ($\text{M} = \text{U}, \text{Np}, \text{Pu}, \text{Am}$) are more common.

5. +6 Oxidation State

The +6 oxidation state is prominent in uranium (U), neptunium (Np), plutonium (Pu), and americium (Am). Uranium forms stable hexafluorides (UF_6) and the uranyl ion (UO_2^{2+}), which is highly stable in aqueous solutions. The stability of the +6 state decreases across the series, with NpO_2^{2+} and PuO_2^{2+} being less stable than UO_2^{2+} .

6. +7 Oxidation State

The +7 oxidation state is rare and observed only in neptunium (Np) and plutonium (Pu). These elements form compounds like NpO_5^{3-} and PuO_5^{3-} under strong oxidizing conditions. The +7 state is highly unstable and exists only in specific environments.

Q.What is meant by spectrochemical series?

[Repeat](#)

Q.What is the position of Lanthanides in periodic table?

[Repeat](#)

Q.Which of the complex ion namely $[\text{Co}(\text{CN})_6]^{3-}$ and $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is an outer orbital complex and why?

[Wrong Question But accurate answer available in above paper , Repeat](#)

Q.What are the shapes of f-orbitals?

Shapes of f-orbitals

For the 'f' orbital $l=3$ then the values of 'm' are $-3, -2, -1, 0, +1, +2, +3$. It shows that the 'f' orbitals have seven orientations as $f_{x(x^2-y^2)}$, $f_{y(x^2-y^2)}$, f_{xyz} , f_z^3 , f_{yz^2} and f_{xz^2} . The

Chemistry of Lanthanides and Actinides 443

f orbital is complicated in shape. The $n=4$, $l=3$ orbitals are called f orbitals, and they're difficult to describe. They have multiple complex features. For example, the $n=4$, $l=3$, $m=0$; $m=1$; and $m=-1$ orbitals are shaped like dumbbells again, but now with two donuts between the ends of the barbell. The other m values look kind of like a bundle of eight balloons, with all their knots tied together in the center.

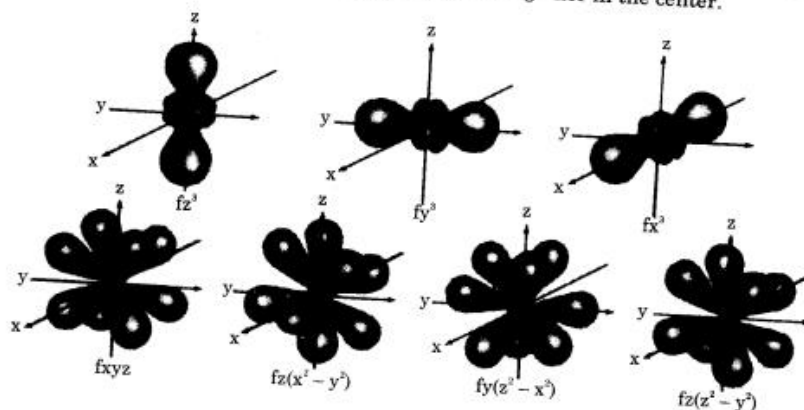


Fig. 7.3: Shapes of f-orbitals.

Vector shapes draw from Google

<https://energywavetheory.com/atoms/f-orbital-shape/>

Q. What is the difference in linkage isomerism and geometrical isomerism?

[Repeat](#)

Q.Explain the magnetic behavior of O₂ molecule by comparing its VBT and MOT pictures?

[Repeat](#)

Q.Give two applications of metal complexes in biological systems?

[Repeat](#)

Q.What is the significance of the magnetism in determining shapes of complexes?

Magnetism plays a key role in understanding the electronic structure of coordination complexes, which helps in predicting their geometry, hybridization, and bonding type.

1. Magnetic Moment Formula

The magnetic moment (μ) helps calculate the number of unpaired electrons using the formula:

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

where **n** is the number of unpaired electrons, and it gives us a direct clue about the complex's electronic structure.

2. Unpaired Electrons Indicate Geometry

If a complex is paramagnetic (with unpaired electrons), it often has high-spin configurations and geometries like tetrahedral or outer-orbital octahedral, while diamagnetic complexes (all electrons paired) usually have square planar or inner-orbital octahedral geometries.

3. Spin State and Ligand Field Strength

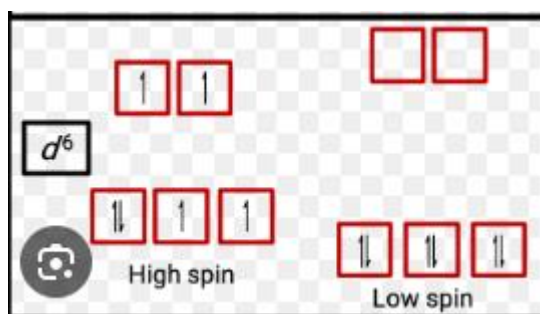
Strong-field ligands like CN^- or CO create large crystal field splitting and form low-spin complexes with fewer unpaired electrons, while weak-field ligands like F^- or H_2O cause small splitting and lead to high-spin complexes with more unpaired electrons.

4. Link with Geometry

The number of unpaired electrons and spin state help determine the shape, for example, d^8 metal ions like Ni^{2+} form square planar complexes with strong ligands (low-spin, diamagnetic) and tetrahedral complexes with weak ligands (high-spin, paramagnetic).

5. Octahedral Complexes Example (d^6 configuration)

For a d^6 octahedral complex, a strong-field ligand will cause electrons to pair in the lower energy orbitals, forming a low-spin (diamagnetic) complex, whereas a weak-field ligand will result in unpaired electrons and a high-spin (paramagnetic) complex.



Q.How CFT is superior to VBT?

Crystal Field Theory (CFT) is considered superior to Valence Bond Theory (VBT) for explaining coordination complexes due to the following key reasons:

Aspect	Valence Bond Theory (VBT)	Crystal Field Theory (CFT)
Explanation of Color	VBT does not explain the colors of coordination compounds, as it ignores the splitting of d-orbitals.	CFT explains color through d-d transitions between split d-orbitals caused by the ligand's electric field, which absorb specific wavelengths of light.
Magnetic Properties	VBT gives a rough idea about magnetic behavior based on	CFT predicts both high-spin and low-spin states depending on ligand strength

Aspect	Valence Bond Theory (VBT)	Crystal Field Theory (CFT)
	unpaired electrons but cannot explain high-spin and low-spin complexes.	and splitting energy, and calculates magnetic moments accurately.
Ligand Strength and Spectrochemical Series	VBT treats all ligands similarly and cannot explain the effect of different ligands on complex stability or spin state.	CFT clearly explains how strong-field and weak-field ligands cause different splitting energies, forming the basis of the spectrochemical series.
Geometry Prediction	VBT assumes geometries based on hybridization (e.g., sp^3 , dsp^2), but fails to explain why certain geometries are favored.	CFT explains geometry based on the pattern of d-orbital splitting (octahedral, tetrahedral, square planar), supported by crystal field stabilization energy (CFSE).

OR

CFT is considered superior to VBT in explaining the properties of coordination complexes due to the following reasons:

Aspect	Valence Bond Theory (VBT)	Crystal Field Theory (CFT)
1. Bonding Description	VBT describes bonding as covalent, formed by the overlap of hybridized orbitals from metal and ligands.	CFT describes bonding as ionic and based on electrostatic interaction between metal ion and ligands.
2. Orbital Concept	It uses hybridization (like sp^3 , dsp^2) to form geometry but doesn't explain energy differences in d-orbitals.	It explains the splitting of d-orbitals in the presence of ligands, which is crucial to electronic structure.
3. Magnetic Properties	Only gives rough predictions by counting unpaired electrons; cannot distinguish high-spin and low-spin cases.	Accurately predicts magnetic behavior and spin states by comparing ligand strength and crystal field splitting.
4. Spectroscopic Properties (Color)	Cannot explain the color of complexes as it lacks treatment of d-d transitions.	Explains colors based on electronic transitions between split d-orbitals absorbing visible light.
5. Spin State Prediction	Limited to hybridization and electron count; fails to explain spin variation in different ligand environments.	Explains high-spin and low-spin states depending on field strength and pairing energy.
6. Geometry Prediction	Assumes ideal hybridization geometries; cannot predict distortions or real-world geometry well.	Predicts actual geometry (octahedral, tetrahedral, square planar) based on d-orbital splitting patterns.
7. Metal-Ligand Interaction	Treats metal-ligand bonding as covalent, mainly via orbital overlap.	Treats it as an electrostatic field that affects metal d-orbitals, explaining many chemical properties.

Q. What is stability constant?

[Repeat](#)

Q.What is the shape of AB₈ molecule according to VSEPR theory?

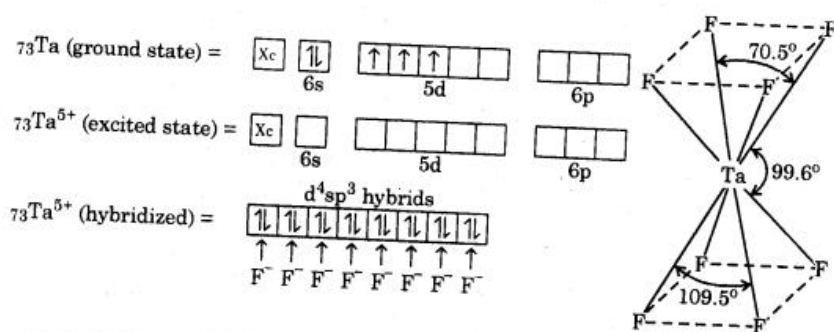
Here A is the central atom and B are the atoms bonded to the central atom with no lone pair of electrons (bp = 8, lp = 0). These bonding electron pairs occupy positions as far apart as possible to minimize electrostatic repulsion. The most stable geometry for AB₈ molecules is **square antiprismatic (a distorted form of dodecahedral)**. In this geometry, the **B—A—B** bond angles vary approximately between **70.5°** and **109.5°** depending on the positions, as shown in the **square antiprismatic structure** of [TaF₈]³⁻.

Examples:

[TaF₈]³⁻ (Tantalum octafluoride complex)

[ReF₈]²⁻ (Rhenium octafluoride complex)

These examples adopt a **square antiprismatic geometry** with sp^3d^4 or $d^4 sp^3$ hybridization, allowing the central atom to form 8 directional bonds with minimal repulsion.



These are two sides of coordination in dodecahedron geometry. Majority of the complexes studied so far of AB₈ type are dodecahedron. The following molecules adopt dodecahedron structure.

Long Questions

Q.What is lanthanide contraction? Why does contraction occur?

[Repeat](#)

Q.Compare the properties of lanthanides and actinides. In which respect they differ from each other?

[Repeat](#)

Q.Discuss shapes of AB₄, AB₂E₂ and AB₃E molecules on the basis of VBT.?

[Haq Nawaz Bhatti Page number 139 onward](#)

Q.Write a note on four electrons three centered bonds?

[Repeat in short](#)

Q.Draw MOT diagram of following coordination compound [Co (NH₃)₆]³⁺ and [CoF₆]³⁻?

[Haq Nawaz Bhatti Page number 292](#)

Q.Write note on colour properties of coordination compound?

[Haq Nawaz Bhatti Page number 274](#)

Beautiful Dua from Quran

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

(2021)



Q. Write the occurrence of actinides in earth crust?

[Haq Nawaz Bhatti page number 479](#)

1. Thorium (Th) and Uranium (U) – The Most Abundant Actinides

Thorium and uranium are the only actinides found in significant quantities in nature. Thorium has an average concentration of $1.6 \times 10^{-3}\%$ in the Earth's crust, while uranium is less abundant at $4 \times 10^{-4}\%$.

- **Uranium** primarily occurs as oxides in minerals like pitchblende (uraninite, U_3O_8), carnotite ($KUO_2VO_4 \cdot 3H_2O$), and autunite ($Ca(UO_2)_2(PO_4)_2 \cdot nH_2O$). Natural uranium consists of three isotopes: ^{238}U (89.27%), ^{235}U (0.72%), and ^{234}U (0.005%). Major uranium-producing countries include Kazakhstan (27.3%), Canada (20.1%), and Australia (15.7%).
- **Thorium** is found in minerals such as thorianite (ThO_2), thorite ($ThSiO_4$), and monazite ($(Th, Ca, Ce)PO_4$). Large deposits exist in the USA (440,000 tonnes), Australia (~300,000 tonnes), and India (~300,000 tonnes).

2. Actinium (Ac) and Protactinium (Pa) – Trace Elements

Actinium and protactinium occur in negligible amounts, primarily associated with uranium ores.

- **Actinium** has an extremely low abundance ($\sim 5 \times 10^{-15}\%$) and is derived from the decay of ^{235}U .
- **Protactinium** is slightly more abundant ($\sim 10^{-12}\%$) and is also linked to ^{235}U decay.

3. Neptunium (Np) and Plutonium (Pu) – Extremely Rare

These elements are present in trace amounts due to their short half-lives.

- **Neptunium** (^{237}Np) exists as an intermediate decay product of uranium but is undetectable in most natural samples.
- **Plutonium** (^{239}Pu) is found in uranium ores at $\sim 10^{-12}\%$ but is primarily synthetic. The longest-lived isotope, ^{244}Pu , has an abundance limit of $3 \times 10^{-20}\%$.

4. Transuranic Elements (Am to Lr) – Synthetic or Nearly Undetectable

Elements beyond plutonium (americium, curium, berkelium, californium, etc.) are **not found naturally** in measurable quantities. They are produced artificially in nuclear reactors or particle accelerators.

Q.What is role of chelates in industry?

Chelates play a vital role in various industries due to their ability to form stable complexes with metal ions, enabling a wide range of practical applications:

Water Treatment: Chelating agents remove or control metal ions in water systems, preventing scale formation and corrosion in boilers, cooling towers, and pipelines. They help soften water by binding calcium and magnesium ions, thus improving industrial water quality.

Agriculture: Chelates enhance nutrient availability by delivering micronutrients like iron, zinc, and manganese to plants in a bioavailable form. They are commonly used in fertilizers to prevent nutrient precipitation and support plant growth in alkaline or calcareous soils.

Catalysis and Chemical Industry: Chelates function as homogeneous catalysts or catalyst stabilizers in various chemical reactions, enhancing efficiency, product yield, and selectivity in industrial processes.

Mining and Metallurgy: In mineral processing, chelating agents facilitate the flotation, extraction, and purification of metals by forming soluble metal complexes, aiding in efficient metal recovery.

Pulp and Paper Industry: Chelates protect bleaching agents from decomposition caused by trace metals, improving bleaching performance, reducing fiber damage, and preventing equipment fouling and discoloration.

Pharmaceutical and Medical Uses: Chelates are used in chelation therapy to treat heavy metal poisoning (e.g., lead, mercury) and as contrast agents in MRI imaging, thanks to their ability to selectively bind metal ions.

Q.Name the factor which effects the stability of metal complexes ?

[Repeat](#)

Q.How Magnetism can be studied?

Magnetism is studied by evaluating the **magnetic properties** of a substance, mainly through **magnetic moment** and **magnetic susceptibility**. These properties help determine whether a substance is **paramagnetic**, **diamagnetic**, or **ferromagnetic**.

1. Magnetic Moment (μ):

- Indicates the number of **unpaired electrons** in an atom or ion.
- For paramagnetic substances, it is calculated using the **spin-only formula**:

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

where:

- n = number of unpaired electrons
- B.M. = Bohr Magneton (unit of magnetic moment)
- **Interpretation:**
 - $\mu > 0 \rightarrow$ **Paramagnetic** (unpaired electrons present)
 - $\mu = 0 \rightarrow$ **Diamagnetic** (no unpaired electrons)

2. Magnetic Susceptibility (χ):

- ❖ Measures the degree of magnetization of a material in an external magnetic field.
- ❖ A positive value indicates attraction (paramagnetism), while a negative value shows repulsion (diamagnetism).

3. Measurement Tools:

- ❖ **Gouy Balance:** Used for measuring the magnetic susceptibility of solids and liquids.
- ❖ **Vibrating Sample Magnetometer (VSM):** Offers high-precision measurements for magnetic moment and behavior.

Applications:

- ❖ Helps determine the electronic configuration of transition metal complexes.
- ❖ Used in identifying the oxidation state and geometry of metal ions in coordination chemistry.
- ❖ Supports analysis of bonding and ligand field effects.

Q.What is the role daily life?

[Haq Nawaz Bhaati Page Number 491](#)

In daily life, actinides have limited but important applications, primarily in:

Smoke Detectors– Americium-241

- ❖ **Americium-241** ionizes air in smoke detectors, creating a current that triggers alarms when disrupted by smoke particles.
- ❖ Its low alpha radiation poses minimal risk, making it safe for household use.
- ❖ These detectors are widely used in homes, offices, and industrial settings for early fire warnings.

Gas Mantles – Thorium-232 (an actinide)

- ❖ Thorium-232 was once used in gas lantern mantles due to its ability to produce intense, steady light when heated.
- ❖ Though efficient, thorium's radioactivity led to its replacement by safer materials like yttrium.
- ❖ Older camping gear and street lamps may still contain traces of thorium-based mantles.

Medical Imaging – Uranium-238

- ❖ Depleted uranium (U-238) shields medical equipment (e.g., X-ray machines) due to its high density and ability to block radiation.
- ❖ It protects patients and technicians from harmful scatter radiation during diagnostics and cancer treatments.
- ❖ Unlike lead, uranium shields are more durable and space-efficient in hospital settings.

Space Power – Plutonium-238

- ❖ powers radioisotope thermoelectric generators (RTGs) in spacecraft (e.g., Mars rovers) by converting decay heat into electricity.

Beyond these, actinides are mostly used in nuclear energy and weapons due to their ability to release massive energy through fission (e.g., uranium-235 in reactors).

Q.Explain the shapes of f-orbitals ?

[Repeat](#)

Q.Discuss shapes of AB₈ and AB₉ molecules on the basis of VBT?

[AB₈ Repeat](#)

AB₉

Here, A is the central atom and B are nine atoms bonded to A, with no lone pairs (bp = 9, lp = 0). To minimize repulsions, the bonding pairs adopt the geometry that gives the most space: a tricapped trigonal prismatic structure. AB₉-type molecules have a tricapped trigonal prismatic geometry with nine ligands arranged around a central atom. Six ligands form a trigonal prism, and three cap its rectangular faces. Bond angles are about 60° within the prism and 90° between prism and capping atoms. The angles vary due to the complex 3D structure.

Examples:

- ❖ **ReH₉²⁻** (Rhenium nonahydride) – A rare AB₉ species.
- ❖ **TcH₉²⁻** (Hypothetical) – Similar structure predicted.

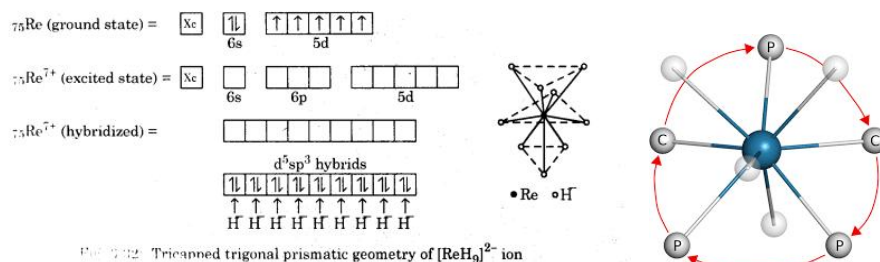


Fig. 5.12. Tricapped trigonal prismatic geometry of $[\text{ReH}_9]^{2-}$ ion

Q.What are Slater and Fajan rules for ionization?

Slater's Rule

The value of σ and hence that of Z_{eff} can be calculated by using Slater's rule. According to these rules the value of σ for a given electron is estimated as follows:

1. Write down the complete electronic configuration of the element and divide the electrons into the following orbital groups starting from the inside of the atom. (1s); (2s, 2p); (3s, 3p); (3d); (4s, 4f); (4d); (4f); (5s, 5p); (6s, 6p) etc.
2. Now select the electron for which the value of σ is to be calculated. For this calculation add up the contributions to σ for the other electrons according to the following rules.

Types of electrons contribution to σ for each electrons of this type.

- | | | |
|-------|--|-------------------------------|
| (i) | All electrons in groups outside the electron chosen. | 0 |
| (ii) | All the electrons in the same group as chosen one | 0.35 or 0.30 for 1s electron. |
| (iii) | All electrons in shell immediately inside. | 0.85 |
| (iv) | All electrons further inside. | 1.00 |
| (v) | There will be no contribution to the value of σ by electrons residing in the orbitals having higher values of their principal quantum number than the orbital containing the electron for which σ is being calculated. | |

Thus σ for an electron residing in ns or np orbital
 $= 0.35 \times (\text{Number of remaining electrons in the nth shell})$
 $+ 0.85(\text{No. of electrons in } (n - 1)\text{th shell})$
 $+ 1.0 (\text{No. of electrons in the inner shell}).$

Example 1.1

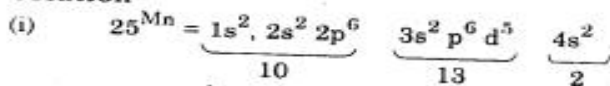
Calculate σ and Z_{eff} for 4s electron in (i) Mn($Z = 28$) (ii) Cu($Z = 29$) and (iii) K($Z = 19$).

Solution

- (i) $^{25}\text{Mn} = \underbrace{1s^2, 2s^2, 2p^6}_{10} \underbrace{3s^2, 3p^6, 3d^5}_{13} \underbrace{4s^2}_2$
 σ for $4s^1$ electron in the Mn atom.

Example 1.1

Calculate σ and Z_{eff} for 4s electron in (i) Mn($Z = 28$) (ii) Cu($Z = 29$) and (iii) K($Z = 19$).

Solution

σ for 4s¹ electron in the Mn atom.

8 Advanced Inorganic Chemistry

$$= 0.35 \times (\text{No. of remaining electron in ns orbital}) \\ + 0.85 \times (\text{No. of electrons in 3s, 3p and 3d orbitals}) \\ + 1.0 (\text{No. of electrons in the inner shell})$$

$$= 0.35 \times 1 + 0.85 \times 13 + 1.0 \times 10 = 21.40.$$

$$Z_{\text{eff}} = Z - \sigma = 25 - 21.40 = 3.60$$

Fajan rules for ionization

[Sanaullah Book Page number 28](#)

OR

<https://www.slideshare.net/slideshow/fajans-rule/231699097>

Q.What are differences between CFT and MOT for explaining metal complex's structure?

Theory	Definition
Molecular Orbital Theory (MOT)	It explains bonding in molecules by combining atomic orbitals to form molecular orbitals that are delocalized over the entire molecule.
Crystal Field Theory (CFT)	It explains bonding in transition metal complexes by considering electrostatic interactions between ligands and metal ion d-orbitals, leading to orbital splitting.

Feature	Molecular Orbital Theory (MOT)	Crystal Field Theory (CFT)
Scope	Applicable to all molecules, including diatomic and polyatomic molecules.	Specifically applicable to transition metal complexes.
Basic Concept	Based on the Linear Combination of Atomic Orbitals (LCAO) to form bonding and antibonding orbitals.	Based on electrostatic interaction between metal ions and ligands that split d-orbitals.
Bonding Explanation	Forms bonding (σ , π) and antibonding (σ^* , π^*) orbitals that affect bond strength.	Explains splitting of d-orbitals into different energy levels (e.g., t _{2g} and e _g).
Orbital Interaction	Molecular orbitals are formed from overlap of atomic orbitals.	Ligands cause repulsion and split the degenerate d-orbitals into different energy levels.

Feature	Molecular Orbital Theory (MOT)	Crystal Field Theory (CFT)
Magnetism	Depends on the presence of unpaired electrons in molecular orbitals.	Depends on high-spin or low-spin configurations caused by weak or strong field ligands.
Color Explanation	Due to electronic transitions between molecular orbitals.	Due to d-d transitions between split d-orbitals absorbing visible light.
Application in Chemistry	Used in predicting bond order, stability, magnetic nature, and behavior of molecules like O ₂ , N ₂ .	Used to explain properties of metal complexes like color, magnetism, and coordination geometry.
Use in Spectroscopy	Useful in UV-Vis and IR spectra based on molecular orbital transitions.	Useful in UV-Vis spectroscopy for d-d transitions and ligand field strength determination.
Energy Diagram	Contains bonding, non-bonding, and antibonding molecular orbitals arranged by energy level.	Shows d-orbital splitting into t _{2g} and e _g in octahedral or other fields with specific Δ values.
Stability Prediction	Bond order = (No. of bonding e ⁻ -- No. of antibonding e ⁻)/2 determines molecular stability.	Crystal Field Stabilization Energy (CFSE) is used to predict the stability of complexes.

<https://imgroupofresearchers.com/molecular-orbital-theory-mot-crystal-field-theory-cft/>

[You can add bonding and anti bonding structure, t_{2g} and e_g structure from book or google](#)

Q.What is the difference between hydrate and linkage isomerism?

[Repeat](#)

Long Questions

Q.How shielding effect alter the properties of lanthanides?

[Available in the portion of lanthanides](#)

Q.Explain the structure of AB₆ types of molecules on the basis of vesp, vbt and mot ?

[According your concept or can find structure from google](#)

Q. Explain jahn teller distortion with example?

[Repeat](#)

Beautiful Dua
from Quran

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

(2020)



Q.Explain John Teller distortion with example?

[Repeat in long](#)

Q.What is the shape of XeF₂ AND XeF₆ molecule?

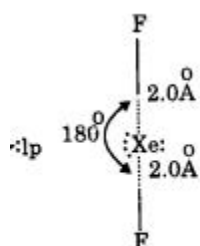


Fig.3.10: Linear geometry of XeF₂ molecule

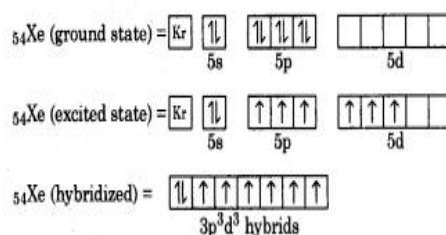
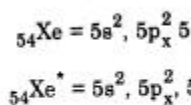


Fig.3.28: Irregular octahedral geometry of XeF₆

Q.How actinides can be separated from lanthanides ?

Repeat

Q. Differentiate inner and outer transition elements?

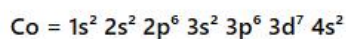
Repeat

Q.What is Fajjan rule?

Repeat

Q.Calculate Z_{eff} for 4s electrons in Cobalt (27)?

Step 1: Write the electronic configuration of Cobalt ($Z = 27$)



Total electrons: 27

We are calculating σ for a 4s electron.

Step 2: Apply Slater's Rules (as shown in image method)

Let's apply the shielding rule for 4s electron:

Grouping the electrons:

- Same group (ns):
1 other electron in 4s
 $\rightarrow 0.35 \times 1 = 0.35$
- (n-1) shell i.e., 3s, 3p, 3d electrons:
 $3s^2 + 3p^6 + 3d^7 = 15$ electrons
 $\rightarrow 0.85 \times 15 = 12.75$
- Inner shell electrons ($1s^2, 2s^2, 2p^6$):
Total = $2 + 2 + 6 = 10$ electrons
 $\rightarrow 1.00 \times 10 = 10.00$

Step 3: Add the shielding contributions

$$\sigma = (0.35 \times 1) + (0.85 \times 15) + (1.00 \times 10)$$

$$\sigma = 0.35 + 12.75 + 10.00 = 23.10$$

Step 4: Use formula to calculate effective nuclear charge (Z_{eff})

$$Z_{\text{eff}} = Z - \sigma = 27 - 23.10 = \boxed{3.90}$$

Final Answer:

$$\boxed{Z_{\text{eff}} \text{ for 4s electron in Cobalt} = 3.90}$$

Q.Differentiate between polarizing power and polarizability?

1.5 POLARIZABILITY AND POLARIZING POWER OF IONS

When a cation, C^+ of an ionic molecule, C^+A^- approach closely the anion, A^- , it withdraws the electrons (i.e. electron cloud) of the anion towards itself and the electron cloud of the anion gets distorted from its symmetrical shape. Thus the electron cloud of the anion no longer remains symmetrical but is elongated towards cation. In other words we say that the A^- anion is distorted (or deformed or polarized) by C^+ cation and this phenomenon is called distortion (or deformation or polarization) of A^- anion by C^+ cation (Fig.1.17).



Fig. 1.17. Schematic representation of polarization of an anion, A^- by a cation, C^+ in C^+A^- ionic molecule (a) No polarization (b) Polarised anion

The cation, C^+ is also deformed by the anion, A^- , but due to smaller size of the cation, its electrons are strongly held to the nucleus and hence the cation is not polarized to an appreciable extent by a nearby anion i.e. the polarization of a cation by an anion is regarded as negligible and hence we generally do not consider the polarization of a cation by anion. The ability of a cation to polarize a nearby anion is called its polarizing power or polarizing ability and the tendency of an anion to get distorted or polarized by a cation is called its polarizability.

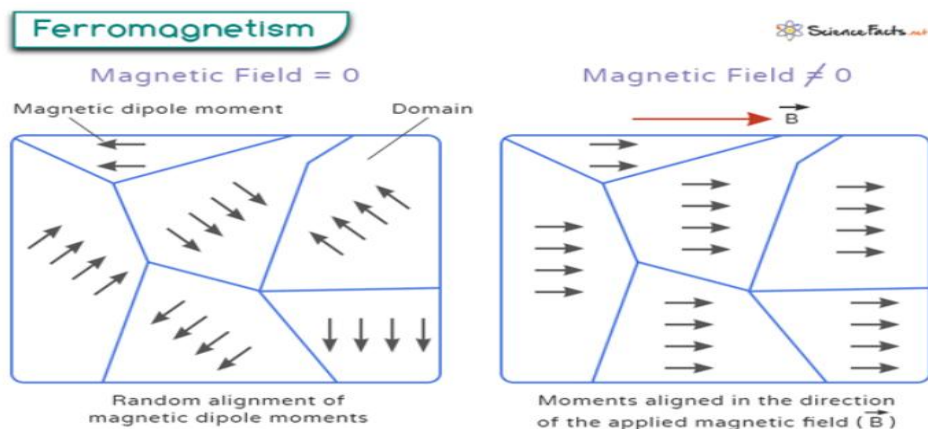
[Sanaullah Book Page Number 28](#)

Q.Define the term Ferromagnetism?

<https://www.hsmagnets.com/blog/ferromagnetic-materials/>

Ferromagnetism is a type of magnetism in which certain materials, such as **iron, cobalt, and nickel**, exhibit **strong and permanent magnetic properties**. This occurs because the **magnetic moments (spins) of unpaired electrons in atoms align parallel to each other** in the same direction, even without the presence of an external magnetic field.

These aligned spins form microscopic regions called **magnetic domains**. In a ferromagnetic material, when an external magnetic field is applied, these domains grow and align in the direction of the field. Importantly, **when the external field is removed, the alignment remains**, giving the material a **permanent magnetism**.



Q. What is stability constant?

Repeat

Q. How CFT is superior to VBT?

Repeat

Long Questions

- i. Write a note on four electrons-three centered bonds?
- ii. What are the role of Lanthanides in industry and chemistry?
- iii. How Thorium can be extracted from monazite ore by acid cracking?
- iv. Discuss shapes of AB_2 , AB_2E and AB_2E_2 Molecules on the basis of it.
- v. How Uranium is extracted from its ores?
- vi. Explain the differences between Lanthanides and Actinides?

All Repeat

Beautiful Dua from Quran

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

(2019)



Q.How VSEPR theory explain the shape of SO₃ and SO₄²⁻?

Here A is the central atom and B are the atoms bonded to the central atom. A species of this type have three bonding electron pairs and no lone pair (bp = 3, lp = 0). These bonding electron pairs occupy their position as far apart as possible so that the electrostatic repulsion between them is minimum. This is most probable if the bond angle between them is 120°. Thus such species have trigonal planar or equivalent triangle, geometry. Examples, BCl₃, BF₃, GaCl₃, BH₃, AlCl₃, SO₃ etc.

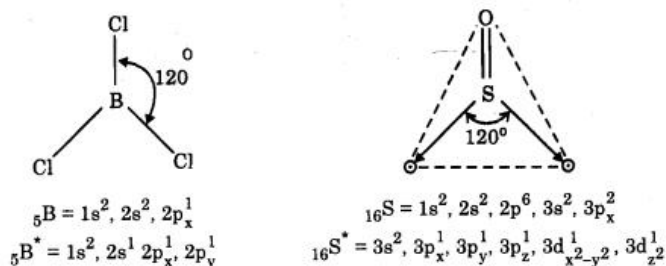


Fig.3.2: Trigonal Planar Geometry of BCl₃ and SO₃ molecules



Sulphate ion

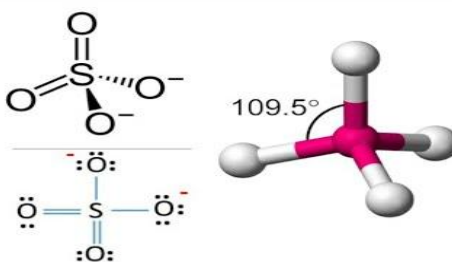
formula:- SO_4^{2-}

Hybridization:- sp^3

Bond angle:- 109.5°

Geometry :- tetrahedral

Shape :- tetrahedral



K2 chemistry class

Q. Explain the shape of AB₃ molecules with examples?

[Repeat](#)

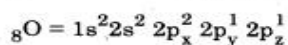
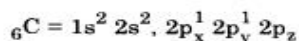
Q. What is the shape of XeF₂ and XeF₄ molecule?

[Repeat](#)

Q. Explain the structure of heteronuclear diatomic molecule by MOT?

1. CO Molecule

The electronic configuration of participating C and O atoms are:



The total number of valence electron is 10 and the bond order would be $\frac{1}{2}(6 - 0) = 3$. This shows that the C and O atoms in CO are bonded by triple bond. There is no unpaired electron in CO molecule, here it is diamagnetic in nature (Fig.3.53). Similarly CN^- and NO^+ have same MO diagram.

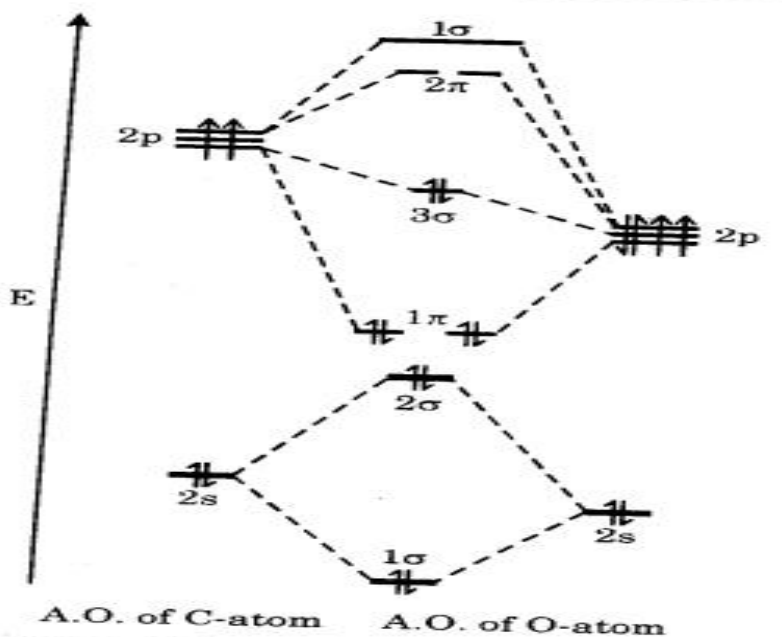


Fig.3.45: Valence shell MO energy levels in CO molecule (only valence rule)

Q.Explain John-Teller distortion with example?

[Repeat](#)

Q.Differentiate between Hybrid and molecular orbital?

Feature	Hybrid Orbital	Molecular Orbital
Definition	A hybrid orbital is formed by the mixing of atomic orbitals within the same atom to form new orbitals of equal energy and specific geometry.	A molecular orbital is formed by the combination of atomic orbitals from different atoms when they bond to form a molecule.
Formation	Formed by intra-atomic mixing of orbitals (s, p, d) during the preparation of an atom for bonding.	Formed by inter-atomic overlapping of orbitals when two atoms come together to form a chemical bond.
Purpose	Explains the geometry and shape of molecules, such as linear, tetrahedral, trigonal planar, etc.	Explains the bonding, bond order, magnetic behavior, and stability of molecules.
Number of Orbitals	The number of hybrid orbitals is equal to the number of atomic orbitals mixed (e.g., mixing 1 s and 2 p gives 3 sp^2).	The number of molecular orbitals equals the total atomic orbitals contributed by the bonded atoms.
Types	Includes types like sp , sp^2 , sp^3 , sp^3d , sp^3d^2 , depending on the combination of orbitals.	Includes bonding, antibonding, and non-bonding orbitals based on constructive or destructive interference of wave functions.
Electron Location	Electrons in hybrid orbitals are localized between two bonded atoms and give rise to directional bonds.	Electrons in molecular orbitals are delocalized over the entire molecule, affecting properties like magnetism and bond order.
Theory Used	Concept used in Valence Bond Theory (VBT).	Concept used in Molecular Orbital Theory (MOT).
Example	In methane (CH_4), carbon forms four sp^3 hybrid orbitals to make four sigma bonds arranged tetrahedrally.	In oxygen molecule (O_2), molecular orbital theory explains paramagnetism due to unpaired electrons in antibonding π^* orbitals.

Q.What are the different oxidation states of Actinides?

[Repeat](#)

Q.Give some examples of ores of Lanthanides.?

[Repeat](#)

Q.What are the applications of Uranium?

1. Nuclear Power Generation

Uranium, especially the isotope Uranium-235, is widely used as a fuel in nuclear power plants. It undergoes nuclear fission, releasing a large amount of energy, which is used to generate electricity.

2. Military Applications

Uranium is used in the production of nuclear weapons. Its ability to sustain a chain reaction makes it a key material in atomic bombs.

3. Depleted Uranium in Ammunition and Armor

Depleted uranium (U-238) is very dense and strong. It is used to make armor-piercing bullets and protective armor for military tanks and vehicles.

4. Medical Use

Uranium reactors are used to produce medical isotopes, such as molybdenum-99, which are used in diagnosing and treating various diseases in nuclear medicine.

5. Scientific Research

Uranium is used in research reactors for experiments in nuclear physics. It provides neutrons and energy for testing and studying atomic behavior.

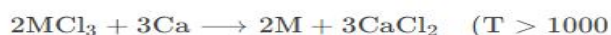
Q.Explain preparation of metals of individual Lanthanide after separation.?

Haq Nawaz Bhaati Page Number 454

Preparation of Individual Lanthanide Metals

1. La, Ce, Pr, Nd, Gd:

- **Method:** Metallothermic reduction of chlorides.
- **Reaction:**



- **Details:** Anhydrous chlorides are reduced with calcium in a tantalum container.

2. Tb, Dy, Ho, Er, Tm, Y:

- **Method:** Metallothermic reduction of fluorides (due to volatility of chlorides).
- **Reaction:**



3. Pm:

- **Method:** Reduction of fluoride with lithium.
- **Reaction:**



4. Eu, Sm, Yb:

- **Method:** Reduction of oxides with lanthanum.
- **Reaction:**



- **Details:** Conducted at 1400 °C in vacuum; volatile metals (Sm, Eu, Yb) are distilled out.

5. General Methods for Mixed Lanthanides (e.g., Misch-metal):

- **Electrolysis of fused chlorides:**
 - $\text{LnCl}_3/\text{NaCl}$ mixture is electrolyzed using graphite electrodes.
 - Metals collect at the cathode and are cast into ingots (not pure).

Long Questions

(i) Explain the isomerism in coordination compounds with suitable examples?

Repeat

(ii) How Thorium can be extracted from monazite ore by acid cracking?

Repeat

(iii) What are the role of Lanthanides in industry and chemistry?

Repeat

**Beautiful Dua
from Quran**

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

(2018)



Q.Why hybrid orbital exhibit better overlap than simple atomic orbitals?

Hybrid orbitals show better overlap than simple atomic orbitals because they are more directional, have a larger lobe, and are geometrically arranged for maximum bonding.

Directionality:

- ❖ Hybrid orbitals like sp^3 , sp^2 , and sp are oriented in specific directions (e.g., tetrahedral, trigonal, linear).
- ❖ This allows them to point directly at the bonding atom, leading to stronger overlap than non-directional s or fixed p orbitals.

Greater Overlap:

- ❖ The shape of hybrid orbitals has one larger lobe, which concentrates electron density in the bonding region.
- ❖ This creates strong sigma (σ) bonds, which are more stable than the bonds formed by simple orbitals.

Explains Equal Bonds & Geometry:

- ❖ In molecules like CH_4 , hybridization forms four equal sp^3 orbitals, explaining the equal bond lengths and 109.5° angle.
- ❖ Without hybridization, the bonds would be unequal and at 90° , which does not match experimental results.

Examples

Methane (CH_4): Carbon uses sp^3 orbitals for four strong, identical σ bonds, explaining the tetrahedral shape.

Ethene (C_2H_4): Carbon atoms use sp^2 hybrids to form strong σ bonds, while the remaining p orbital forms a π bond.

BeCl_2 : The beryllium atom forms two sp hybrid orbitals to bond symmetrically with chlorine atoms in a linear shape.

[You have Ability , draw these structures.](#)

Q.Differentiate inner and outer transition elements.?

[Repeat](#)

Q.What is Jahn Teller distortion theorem?

[Repeat](#)

Q.Differentiate between polarizing power and polarizability?

[Repeat](#)

Q.What is Fajjan rule?

[Repeat](#)

Q.What are post transition elements? Give examples

Post-transition elements are metallic elements found in the p-block of the periodic table, typically in Groups 13 to 16. They come after the transition metals and exhibit distinct physical and chemical properties. These elements generally have **lower melting and boiling points**, are **softer**, and possess **higher electronegativities** compared to transition metals. Although they can show **variable oxidation states**, this behavior is less pronounced than in transition metals.

Unlike transition metals, post-transition elements **do not have partially filled d-orbitals**, which makes them **less effective as catalysts** and gives them different bonding characteristics. Many of them form **amphoteric oxides** and act as **Lewis acids** in chemical reactions.

Examples of post-transition elements

It include **aluminum (Al)**, **gallium (Ga)**, **indium (In)**, **tin (Sn)**, **lead (Pb)**, and **bismuth (Bi)**. These elements are widely used in daily life—for instance, aluminum in packaging and construction, and lead in batteries.

Q.Calculate Z_{eff} for 4s electrons in Cobalt (Z=27)?

Repeat

Q.What are hydrate isomers? Give examples.

Repeat

Q.Can lanthanum ion exist in + 4 oxidation state? If yes then how, if no then why?

No, lanthanum ion (La) cannot stably exist in the +4 oxidation state under normal conditions.

Lanthanum belongs to the **lanthanide series** and has the electron configuration:

La: [Xe] 5d¹ 6s²

In chemical reactions, lanthanum **readily loses three electrons** (5d¹ and 6s²) to form the **+3 oxidation state (La³⁺)**, which is **highly stable** due to a noble gas core ([Xe]) configuration.

To form the +4 state (La⁴⁺), lanthanum would have to **remove one more electron from a completely filled inner shell**, which requires a very high amount of energy and is **not energetically favorable**. Also, La⁴⁺ would be **highly unstable and strongly oxidizing**, and thus not found in normal chemistry.

Therefore, **lanthanum shows only the +3 oxidation state**, and **+4 oxidation state is not possible** due to **electronic configuration and instability**.

Q.What is 4c-3e bond? Give example

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Long Question

Q.What is isomerism? Discuss its different types exhibited by coordination Compound?

Repeat

Q.4. What is lanthanide contraction? Discuss the phenomenon in detail along with its consequences?

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Q.5. What are main postulates of VSEPR theory? Discuss shapes of AB₄, AB₃E, AB₂E, molecules on the basis of it?

Haq Nawaz Bhatti Page number 136 Onward

**Beautiful Dua
from Quran**

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

(2017)



Q. Write down two main uses of Lanthanides?

1. Production of Alloys

Lanthanides are added to metals to form strong, corrosion-resistant alloys. These alloys are useful in making aircraft engine parts and nuclear reactors due to their ability to withstand high temperatures. Mischmetal, a common alloy of lanthanides, is used in lighter flints and special steels.

2. Phosphors and Lighting

Lanthanide compounds are used in making phosphors for LED lights, TV screens, and fluorescent lamps. Elements like europium and terbium produce bright colors when exposed to energy. Their glowing ability helps improve display quality in electronic devices.

Q.Mention any two applications of inorganic compounds in biological system?

1. Hemoglobin (Iron – Fe)

Iron (Fe) is an essential inorganic element in the structure of hemoglobin, a protein found in red blood cells. It binds with oxygen and helps transport it from the lungs to the rest of the body. Without iron, the body cannot produce enough healthy red blood cells, leading to fatigue and weakness.

2. Bones and Teeth (Calcium Phosphate – $\text{Ca}_3(\text{PO}_4)_2$)

Calcium phosphate is a key inorganic compound found in bones and teeth. It provides strength and structure to the skeletal system. This compound ensures bone density and supports essential biological functions like muscle contraction and nerve signaling.

Q.Briefly narrate the occurrence of actinides in earth crust?

[Repeat](#)

Q.What are inner transition elements?

[Repeat](#)

Q.Calculate Zeff for 4s electrons in Cobalt (Z=27).?

[Repeat](#)

Q.What is hybridization in XeF_2 molecule in solid state?

[Repeat](#)

Q.How CFT is superior to VBT?

[Repeat](#)

Q.Draw the structures for following systems as per VSEPR model

1) AB_4E_2

ii) AB_3E_3

[Repeat](#)

Q.Define the term Ferromagnetism?

[Repeat](#)

Q.What is stability constant?

[Repeat](#)

Long Question

Q. Discuss and compare the structures of following coordination compounds on the basis of MOT.



Haq Nawaz Bhatti page number 292 Same concept applicable for both ,water weak and cyanide strong

Q. What is lanthanide contraction? Discuss the phenomenon in detail along with its consequences?

Repeat

a) Briefly discuss Jahn Teller distortion Theorem?

Repeat

b). Write a note on four electrons-three centred bonds?

Repeat

Q.Explain how the MOT is used to predict the spectroscopic and magnetic properties transition metal coordination compounds?

Molecular Orbital Theory (MOT) is a quantum mechanical approach that explains the bonding, structure, and properties of coordination compounds by considering the interaction between the metal's atomic orbitals and the ligand orbitals to form molecular orbitals. Unlike Crystal Field Theory (CFT), which treats metal-ligand interaction as purely electrostatic, MOT accounts for **covalent interactions**, including both σ (sigma) and π (pi) bonding. This makes it a powerful tool to predict **spectroscopic behavior (color, transitions)** and **magnetic behavior (paramagnetism or diamagnetism)** of transition metal complexes.

1. Spectroscopic Properties

MOT explains the color and absorption behavior of coordination compounds by showing how metal and ligand orbitals combine to form molecular orbitals with different energy levels. In octahedral complexes, metal d, s, and p orbitals interact with ligand orbitals to form bonding, non-bonding, and antibonding orbitals. The d-orbitals are split into two sets: the lower-energy t_{2g} orbitals and the higher-energy antibonding e_g^* orbitals. When visible light is absorbed, electrons get excited from t_{2g} to e_g^* , producing a **d–d transition**. The energy gap between these orbitals, known as Δ_o , determines the wavelength of light absorbed and the color observed.

MOT also explains intense **charge transfer transitions**, where electrons move between metal and ligand orbitals. In **Ligand-to-Metal Charge Transfer (LMCT)**, electrons shift from ligand to metal orbitals, while in **Metal-to-Ligand Charge Transfer (MLCT)**, electrons move from metal to ligand π^* orbitals. These transitions are more intense and allowed by selection rules.

Example: The complex $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ appears purple because it absorbs green light due to a weak d–d transition between metal-centered orbitals. In another case, the **permanganate ion** $[\text{MnO}_4]^-$ is deep purple due to a strong LMCT transition from the oxygen ligands to the Mn^{7+} metal center. Similarly, in $[\text{Ru}(\text{bpy})_3]^{2+}$, an MLCT transition from Ru^{2+} to the π^* orbital of the bipyridine ligands gives the complex an intense orange color. These examples show how MOT helps explain not only weak d–d transitions but also strong charge transfer bands.

2. Magnetic Properties

MOT describes magnetic behavior by analyzing how electrons are filled into molecular orbitals formed from metal and ligand orbitals. According to the Aufbau principle, Hund's rule, and Pauli exclusion principle, electrons occupy orbitals based on energy levels and pairing rules. If the ligand field is weak (low Δ_o), electrons prefer to remain unpaired in higher orbitals, forming **high-spin complexes** that exhibit **paramagnetism** due to unpaired electrons. In contrast, strong-field ligands cause larger splitting (high Δ_o), so electrons pair up in the lower orbitals, forming **low-spin complexes** that are **diamagnetic**, having no unpaired electrons.

The magnetic moment of a complex is related to the number of unpaired electrons and is calculated using the formula:

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

where n is the number of unpaired electrons. This helps quantify the paramagnetism of a complex.

Example:

The complex $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ is high-spin because H_2O is a weak-field ligand. The iron (Fe^{2+} , d^6) has four unpaired electrons, resulting in a magnetic moment of about 4.9 B.M., indicating paramagnetism. On the other hand, in the complex $[\text{Fe}(\text{CN})_6]^{4-}$, CN^- is a strong-field ligand, and the same d^6 metal ion forms a low-spin complex with all electrons paired, giving a magnetic moment of 0 B.M., making it diamagnetic. This contrast clearly shows how MOT helps predict magnetic behavior based on electron configuration and ligand strength.